

SIMULATION AND PERFORMANCE EVALUATION OF COMPUTER NETWORKS USING NETSIM

COMPUTER NETWORKS(TH-E2+TE2)



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PREFACE

The field of Computer Networks plays a vital role in modern communication systems, forming the foundation of data exchange, internet services, and real-time applications across the world. To truly understand how networks function, it is essential to go beyond theoretical concepts and observe the behavior of network protocols in controlled environments.

This project, carried out using NetSim Lite, was designed to provide practical exposure to the working principles of network protocols such as TCP, UDP, FTP, HTTP, and VoIP, as well as to analyze concepts like packet loss, congestion, delay, and routing. The experiments conducted simulate real-world network scenarios and help visualize how data flows between connected devices under different conditions.

Through these experiments, we explored the Transport, Application, and Network layers of the OSI model, understanding how reliability, speed, and efficiency are managed in various protocols. For instance, TCP's reliability and congestion control mechanisms were compared with UDP's lightweight, connectionless approach. Similarly, routing algorithms like RIP and OSPF were studied to observe how data finds the most efficient path in dynamic networks.

NetSim Lite, a professional-grade network simulator, served as the primary tool for all simulations. It allowed us to design topologies, configure nodes, vary parameters such as bandwidth, delay, and packet error rate, and monitor real-time performance metrics like throughput, jitter, and packet loss. The platform's visualization and result analysis features made it easier to comprehend complex networking behaviors.

This project helped bridge the gap between theoretical understanding and real-world networking. It deepened our appreciation of how protocols cooperate to ensure reliable communication and how network conditions such as congestion or delay impact performance.

In summary, this work aims to present a comprehensive study of key computer networking concepts through practical experimentation using NetSim Lite — reinforcing the fundamentals of communication, reliability, and efficiency in modern networks.

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Experiment 1: TCP File Transfer

Aim

To study the performance of file transfer using TCP and understand how reliability and congestion control affect throughput and delay.

Introduction

TCP (Transmission Control Protocol) is a **connection-oriented** transport protocol that provides reliable, ordered, and error-free delivery of data. It is widely used for file transfer applications where data accuracy is critical. However, due to acknowledgments and retransmissions, TCP may experience delays under high latency or loss conditions.

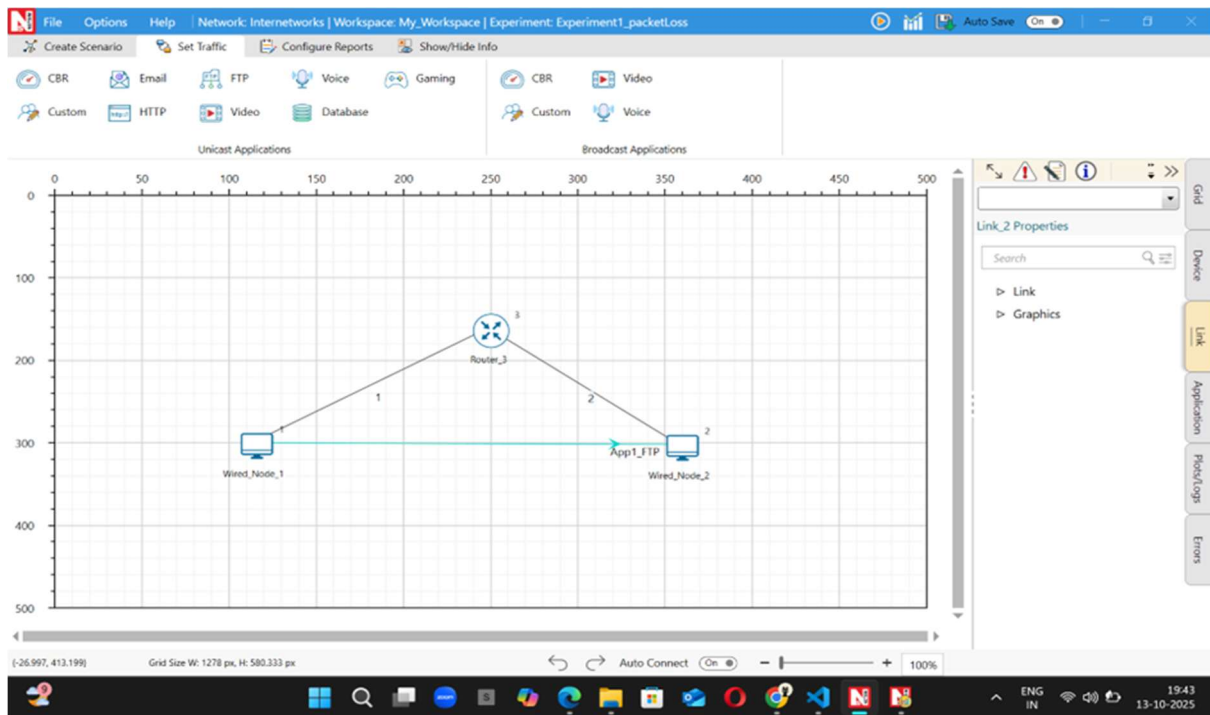
Theory

TCP ensures reliable delivery through sequence numbers, acknowledgments, and retransmissions. It uses **flow control** (via sliding window) and **congestion control** algorithms like Slow Start and Congestion Avoidance to adapt to network conditions.

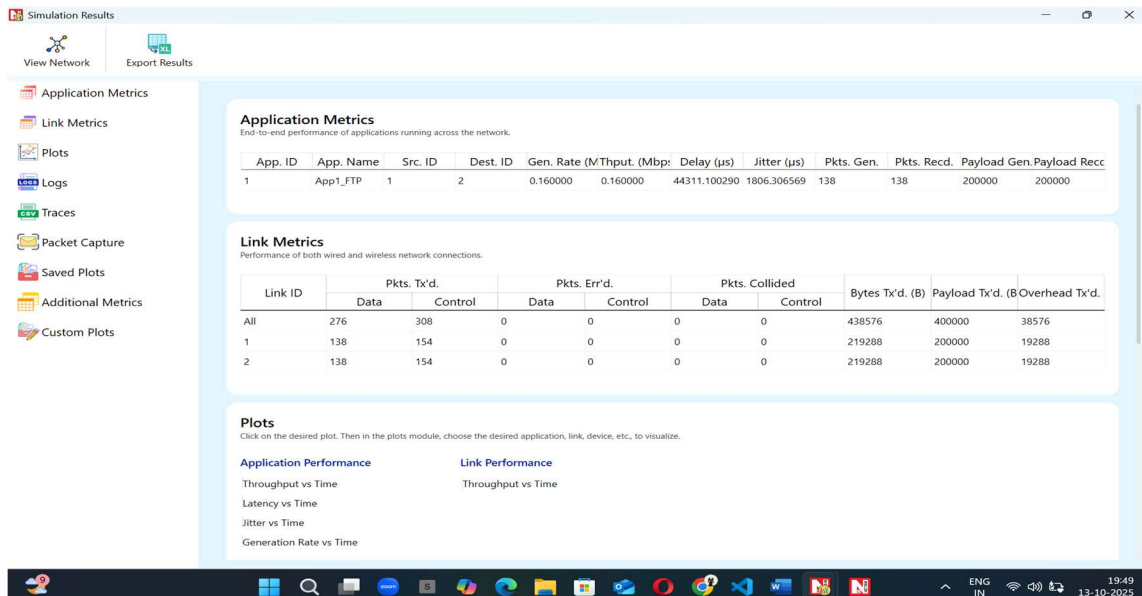
Setup / Procedure

1. Create a topology with a TCP client and server in NetSim Lite.
2. Configure link bandwidth, propagation delay, and file size.
3. Run simulations for different file sizes and delays.
4. Note throughput, delay, and transfer time.

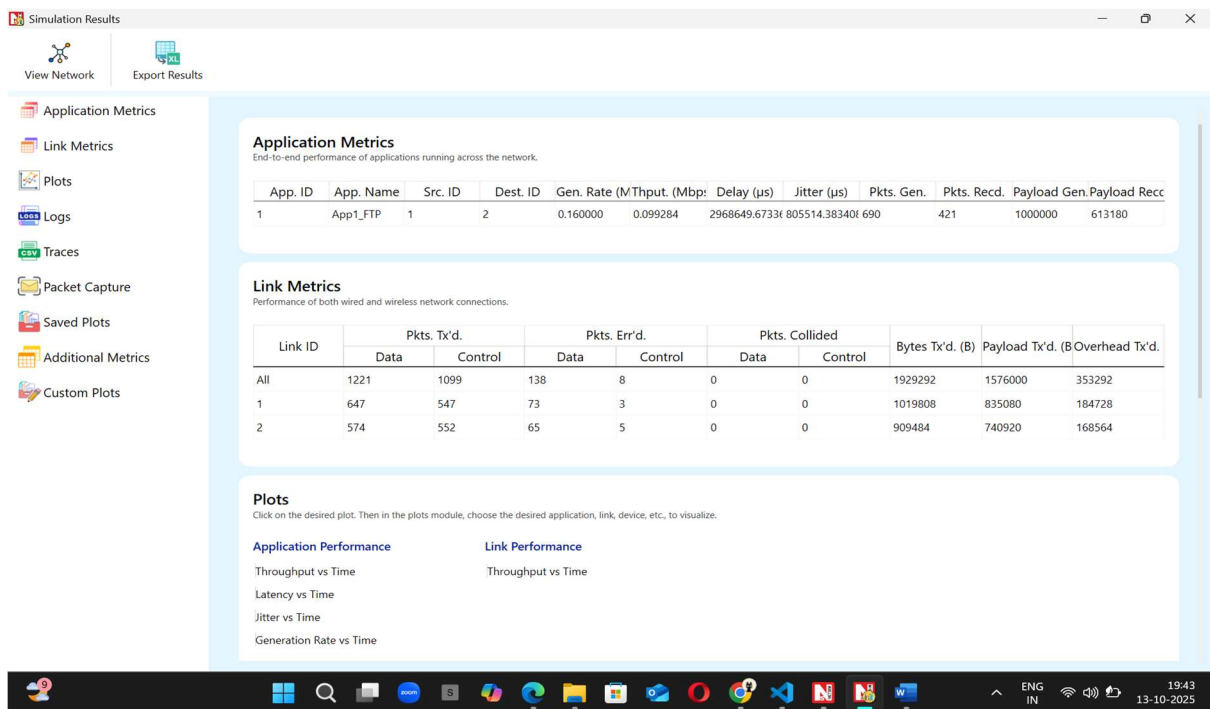
TOPOLOGY:



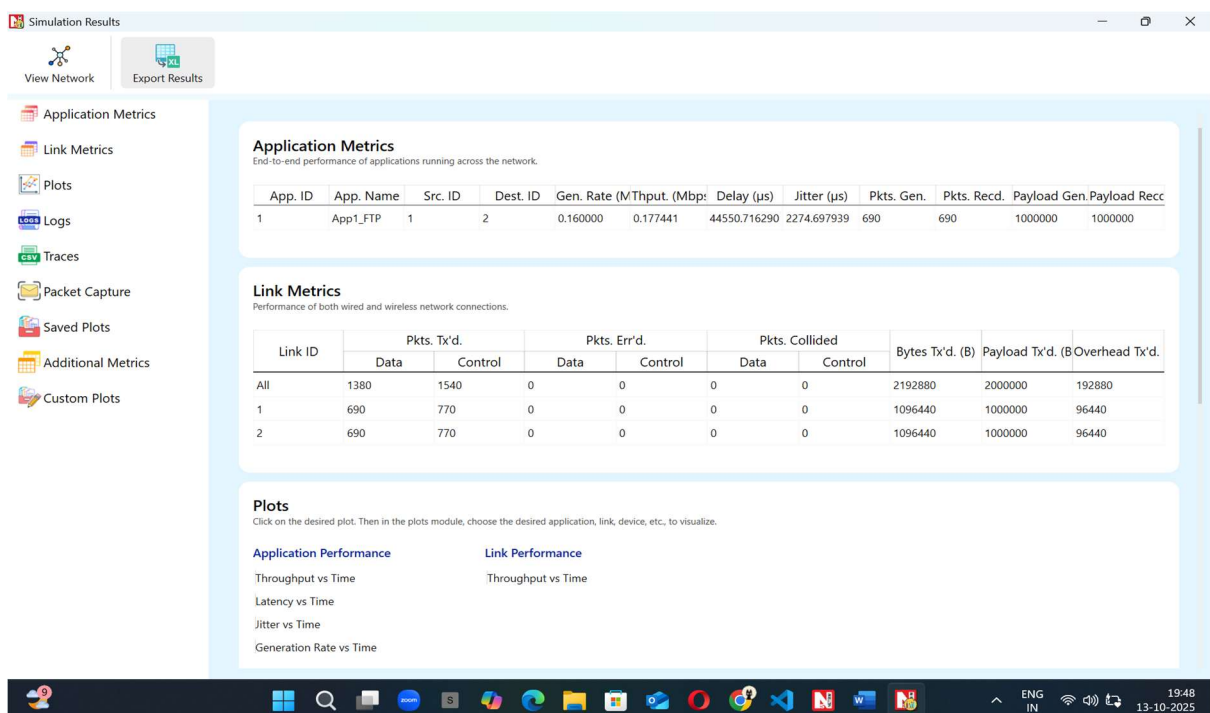
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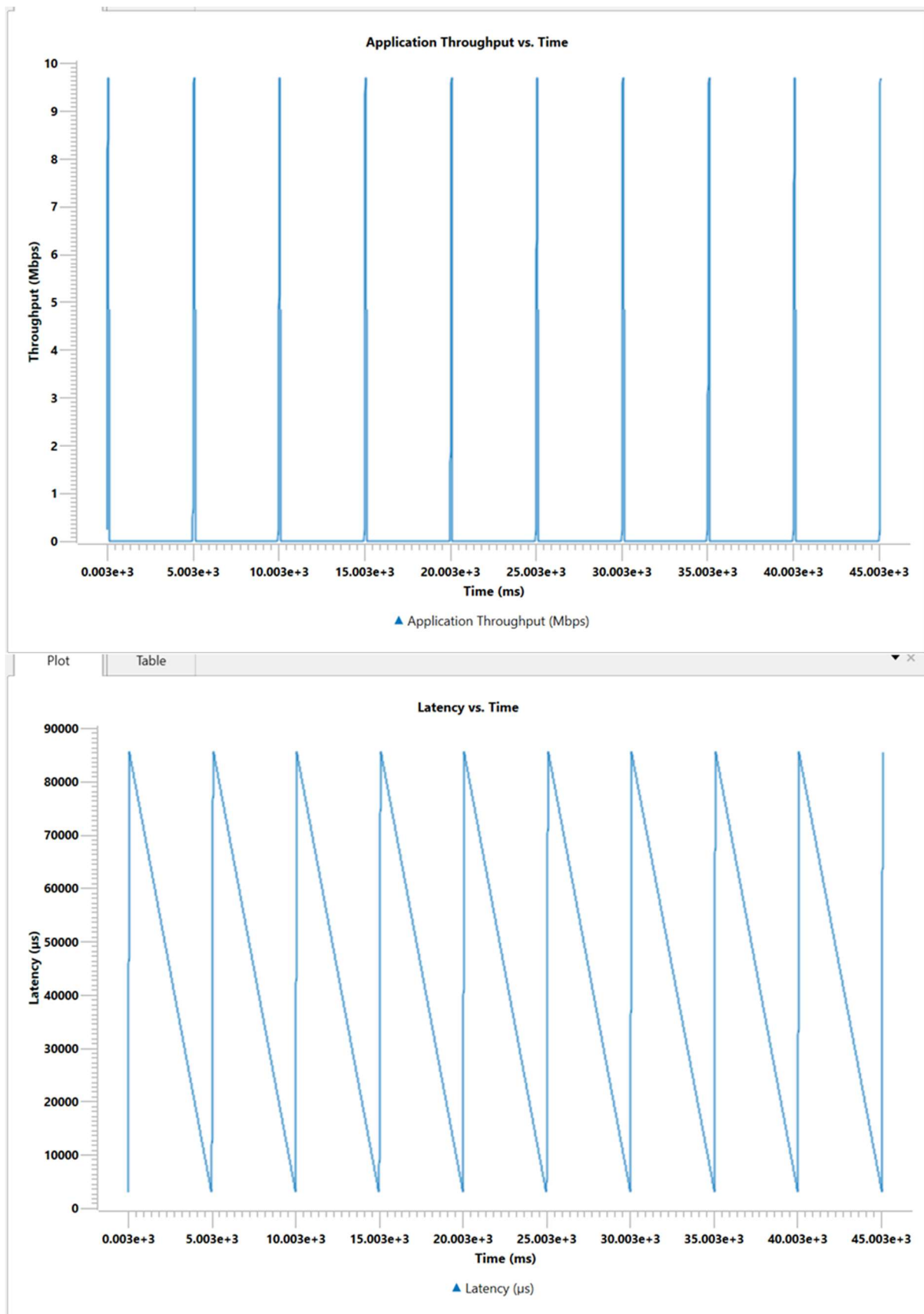
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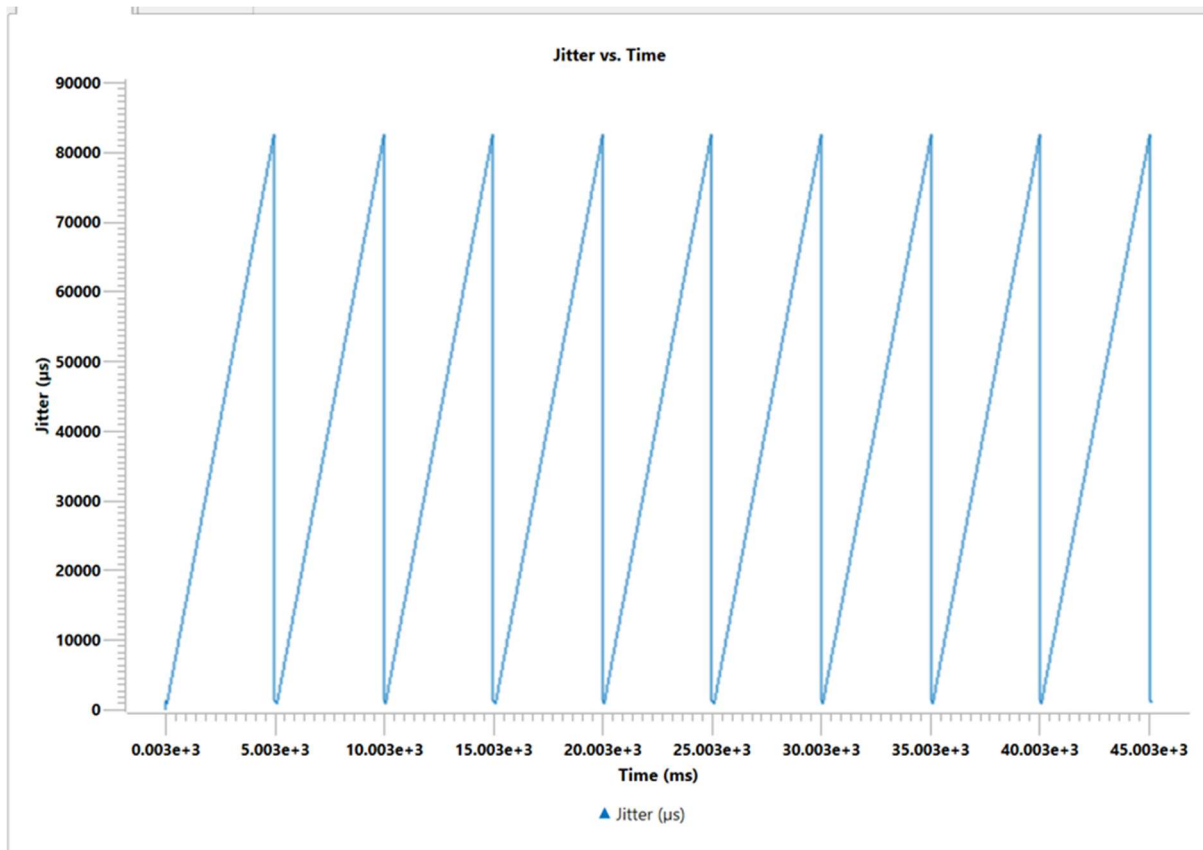


TESTCASE 3:



GRAPHS:





Observation

Throughput decreases with higher delay and larger file size due to acknowledgment waiting time.

Conclusion

TCP provides reliable delivery but its performance decreases in high-delay or congested networks.

Experiment 2: UDP File Transfer

AIM

To study the behaviour of file transfer using the UDP protocol and compare it with TCP.

INTRODUCTION

UDP (User Datagram Protocol) is a **connectionless** transport protocol that sends data without acknowledgments or retransmissions. It provides faster data transfer but without reliability, making it ideal for real-time applications.

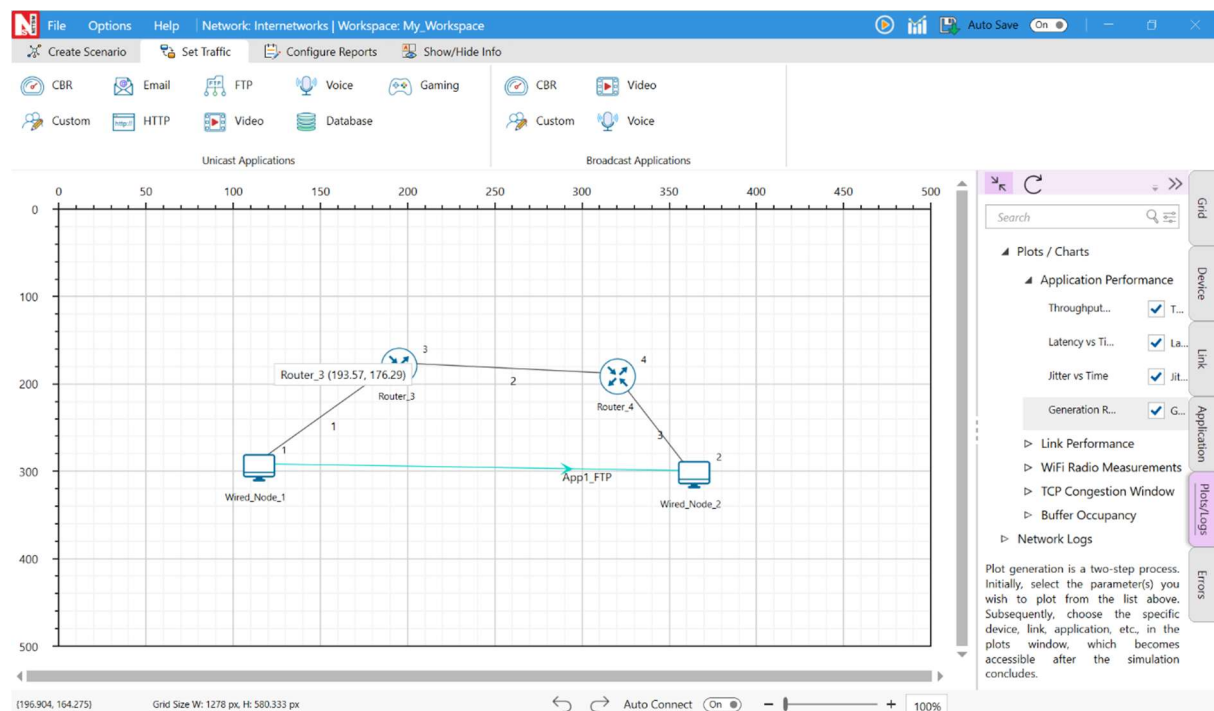
THEORY

UDP transmits packets independently, without establishing a connection. It lacks flow and congestion control, so packet loss may occur if the network is overloaded.

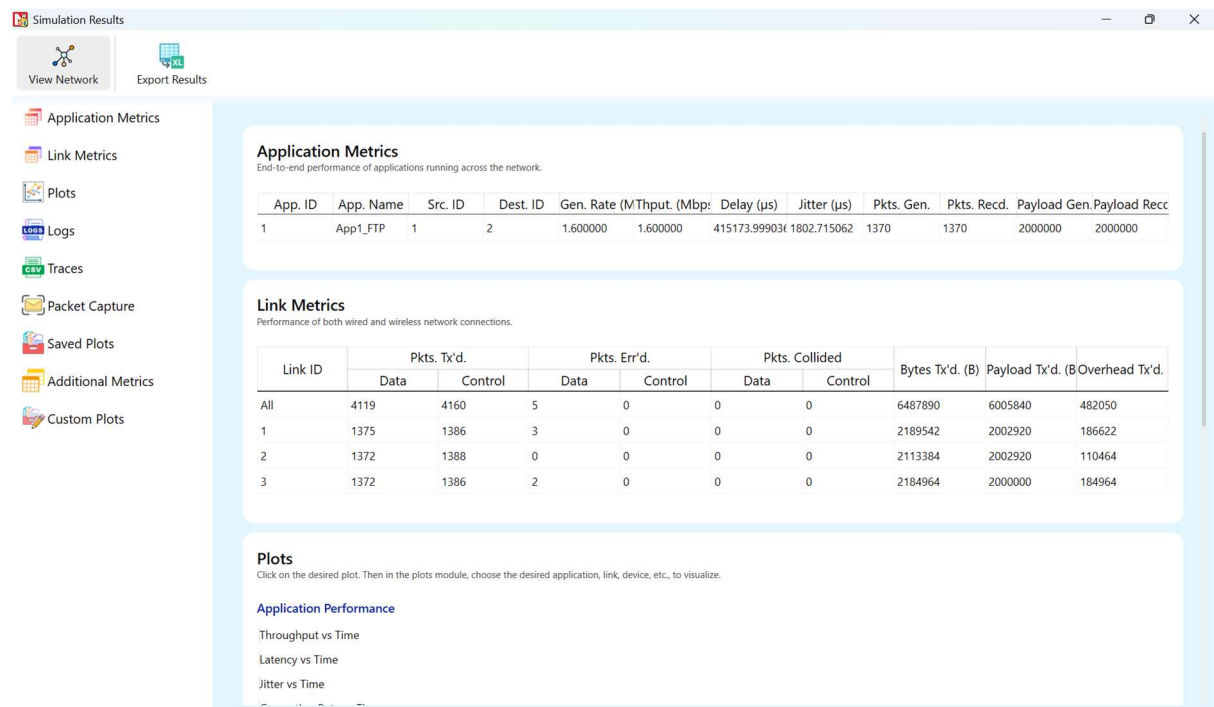
SETUP / PROCEDURE

1. Set up a UDP client and server in NetSim Lite.
2. Configure bandwidth, delay, and packet size.
3. Run tests under different network loads.
4. Record throughput, delay, and packet loss.

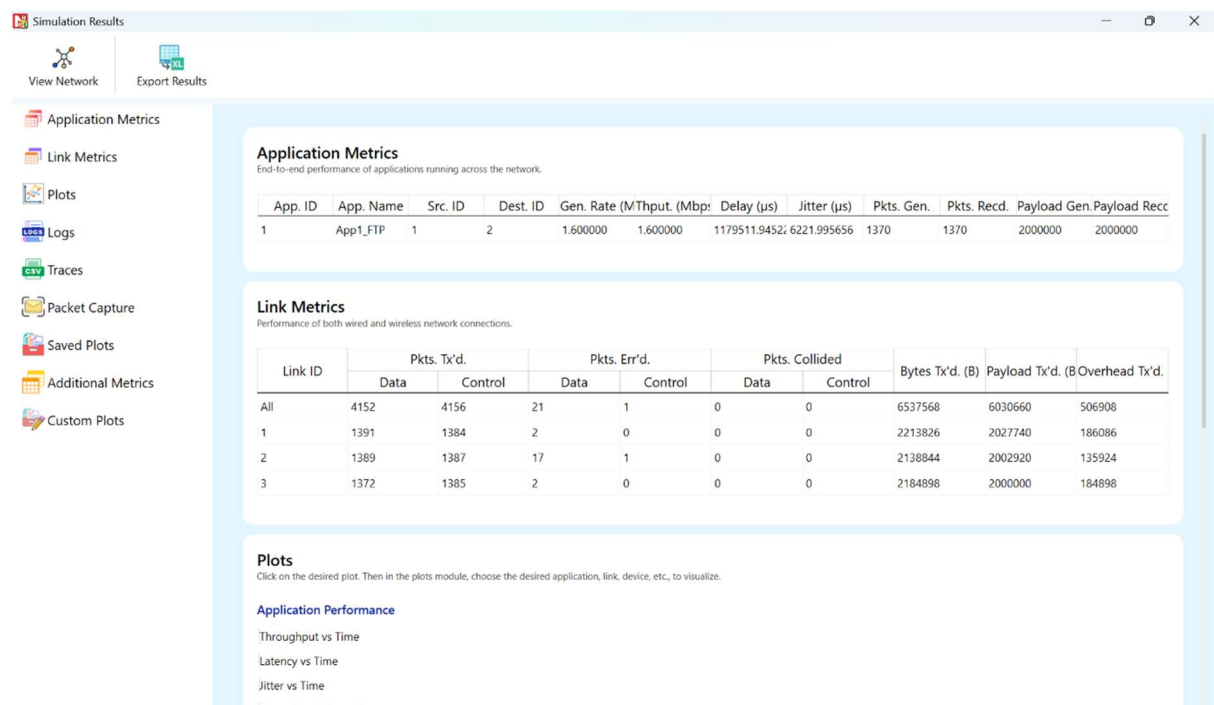
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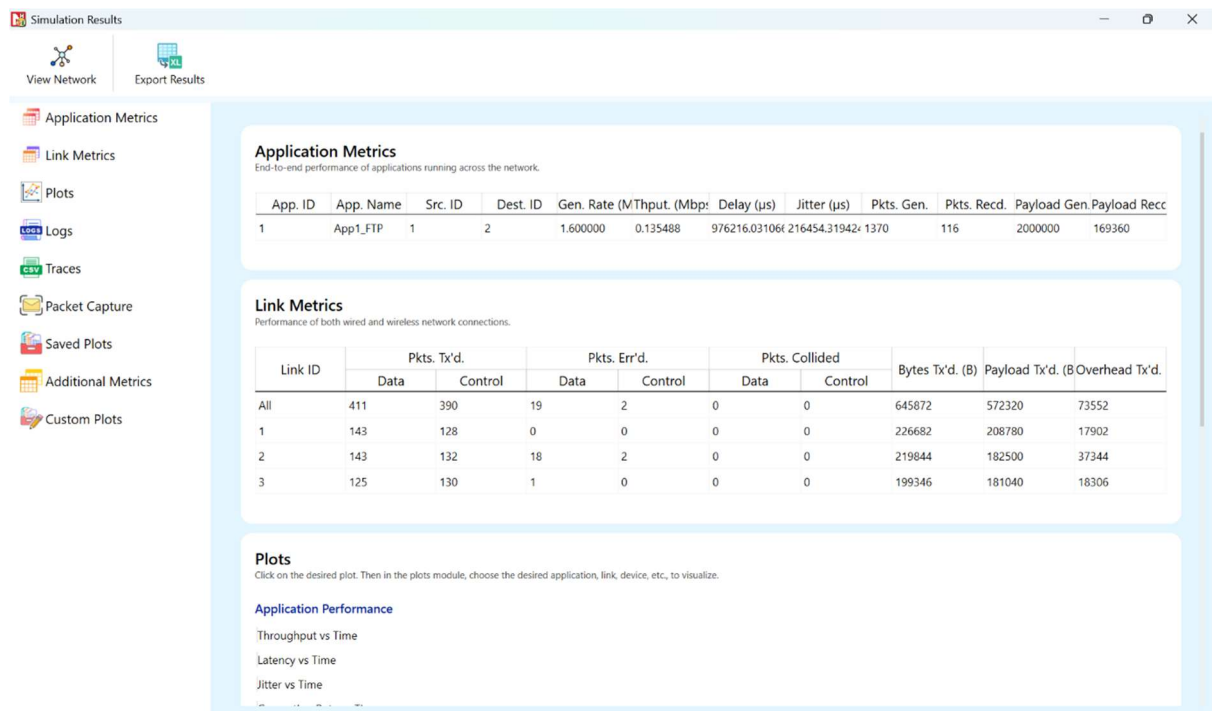
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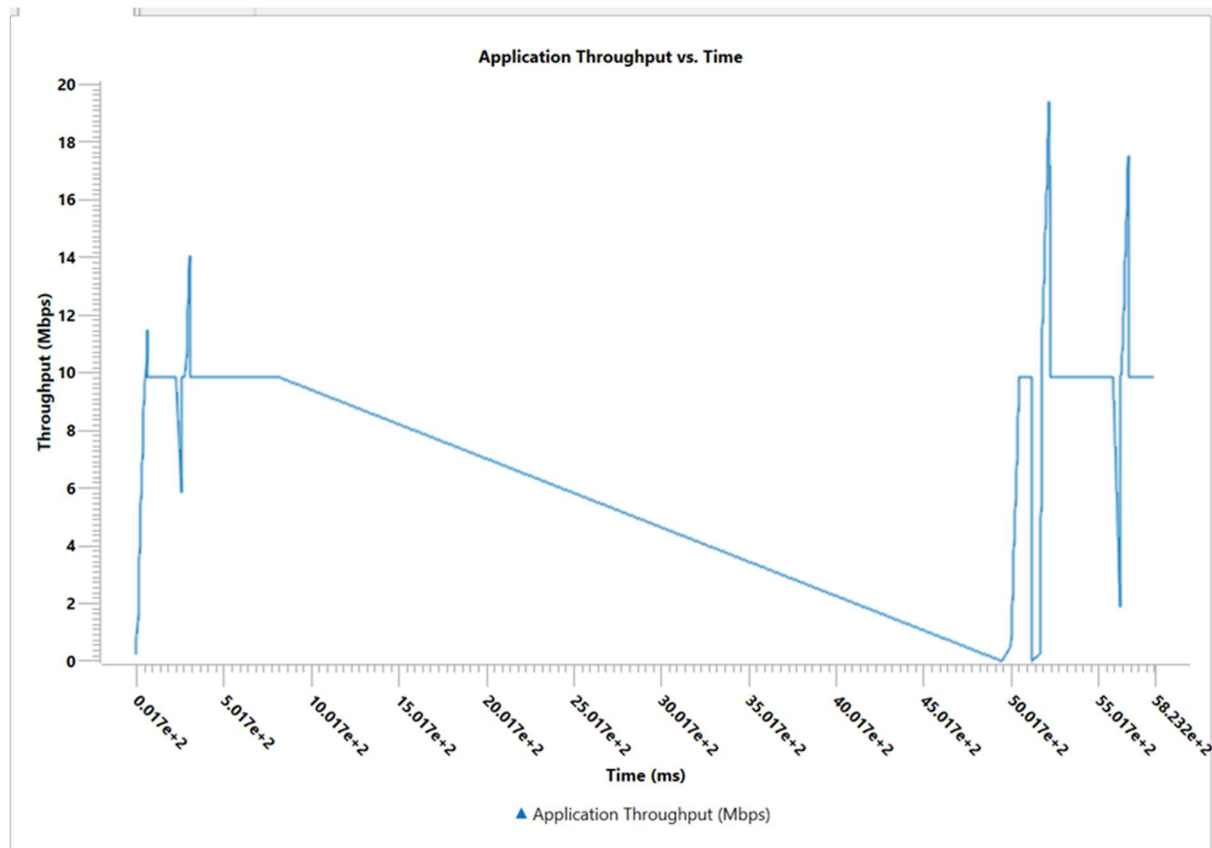
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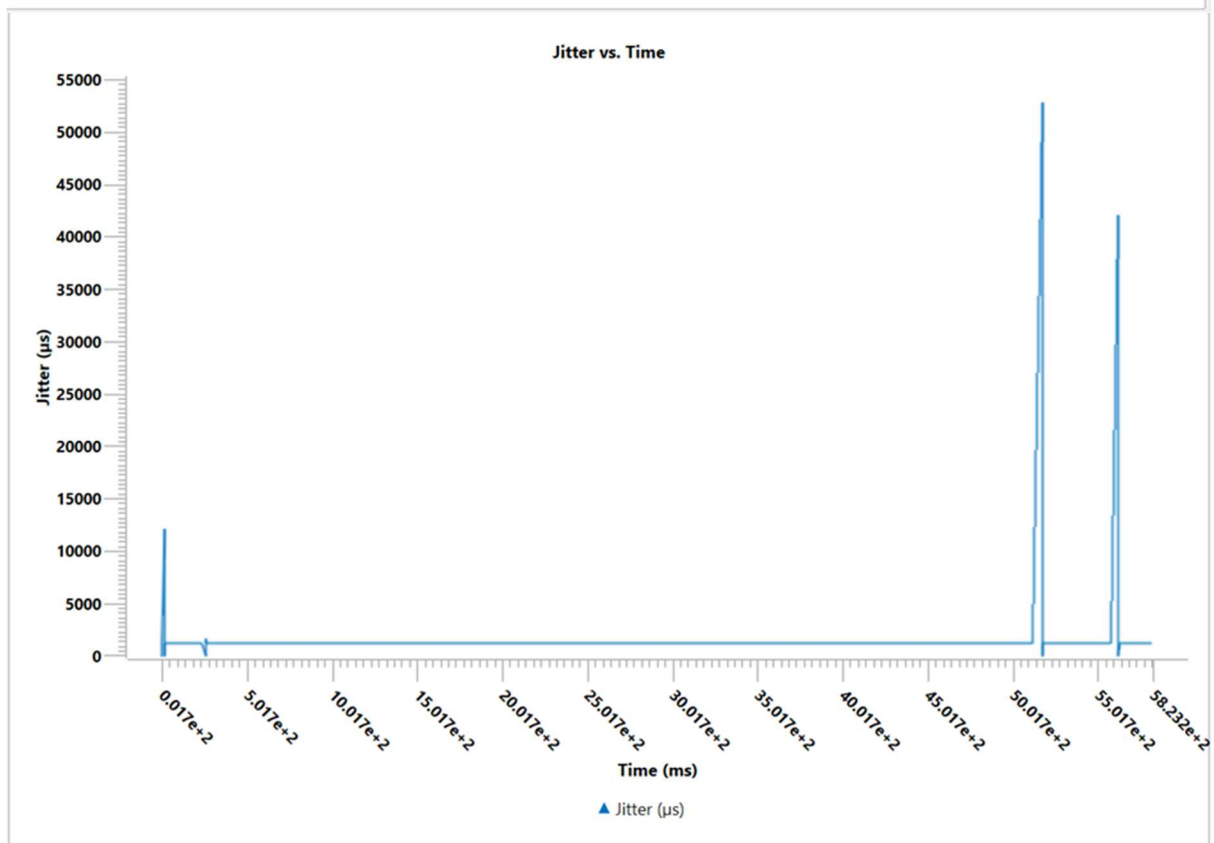
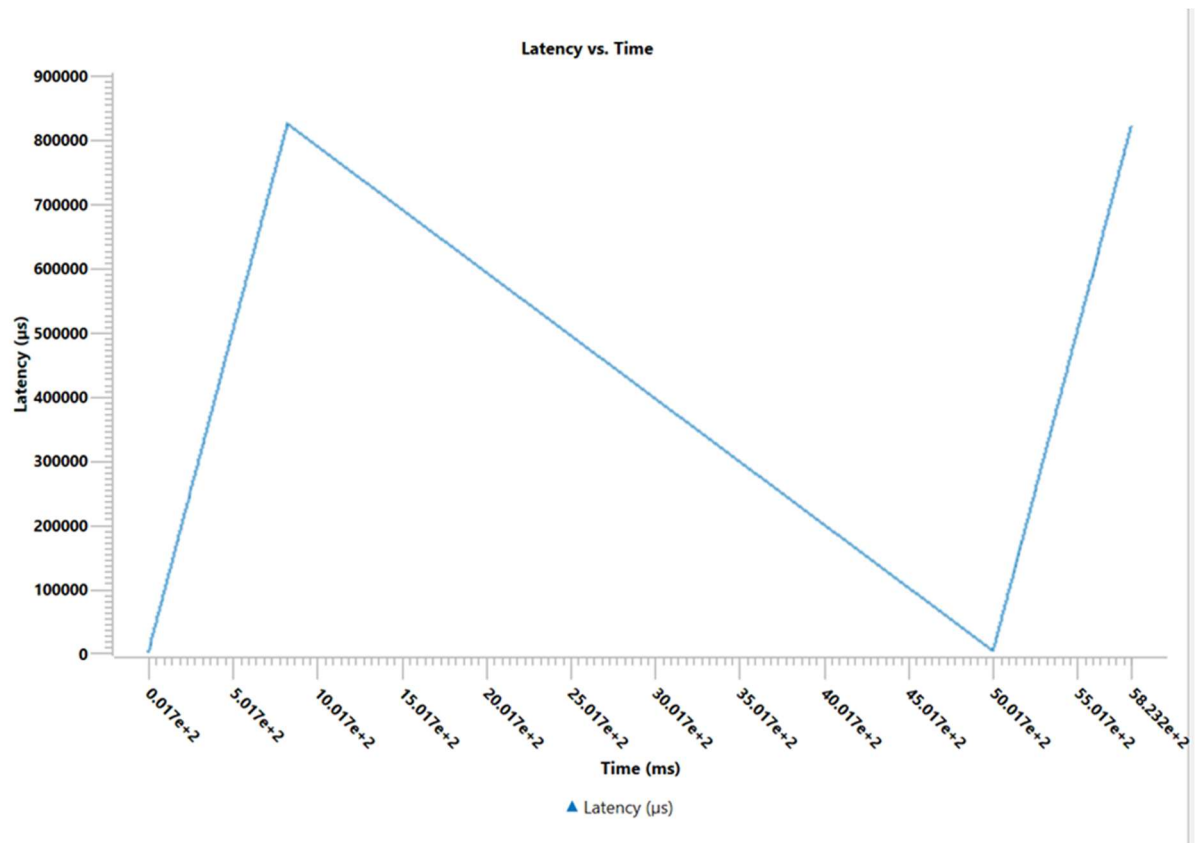


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OBSERVATION

UDP achieves higher throughput than TCP but may lose packets in congested networks.

CONCLUSION

UDP offers faster transmission but sacrifices reliability, making it suitable for delay-sensitive data transfer.

Experiment 3: TCP under Packet Loss

AIM

To analyse TCP performance when packet loss occurs in the network.

INTRODUCTION

TCP uses retransmissions to recover from lost packets. While this maintains reliability, it reduces throughput and increases delay.

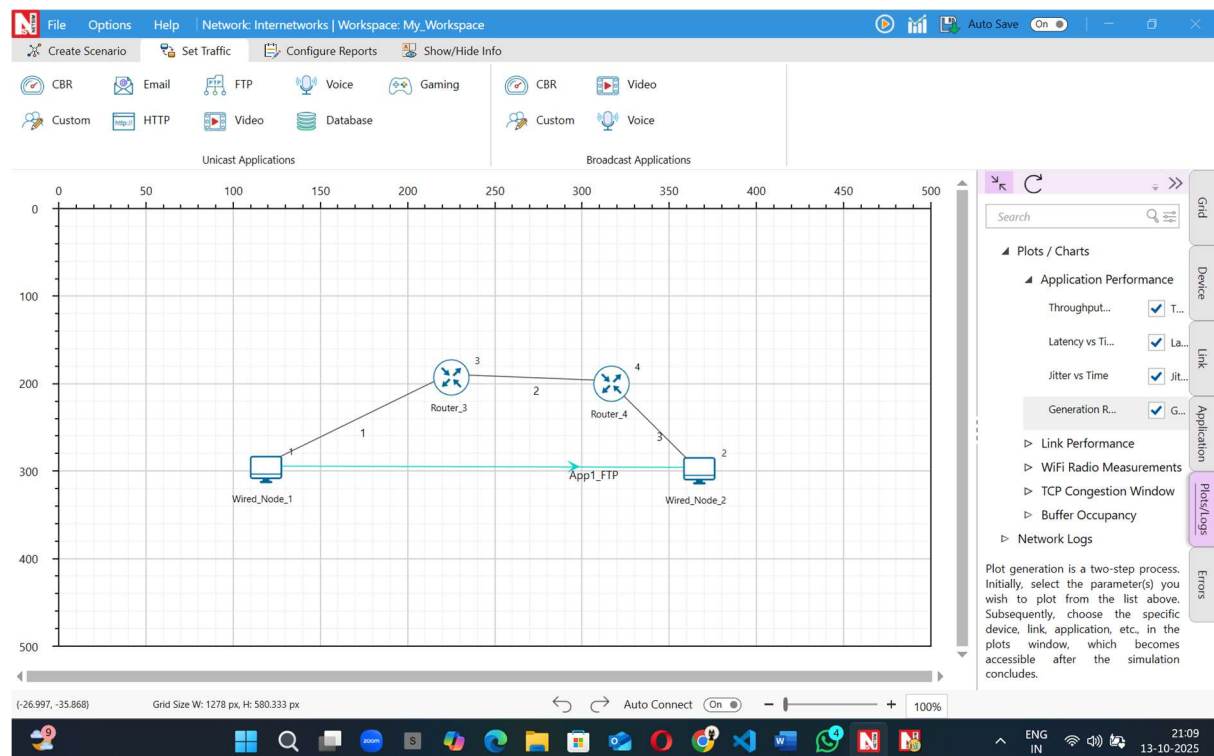
THEORY

When packet loss occurs, TCP triggers retransmissions and enters congestion control phases. Increased packet loss causes TCP to slow its sending rate.

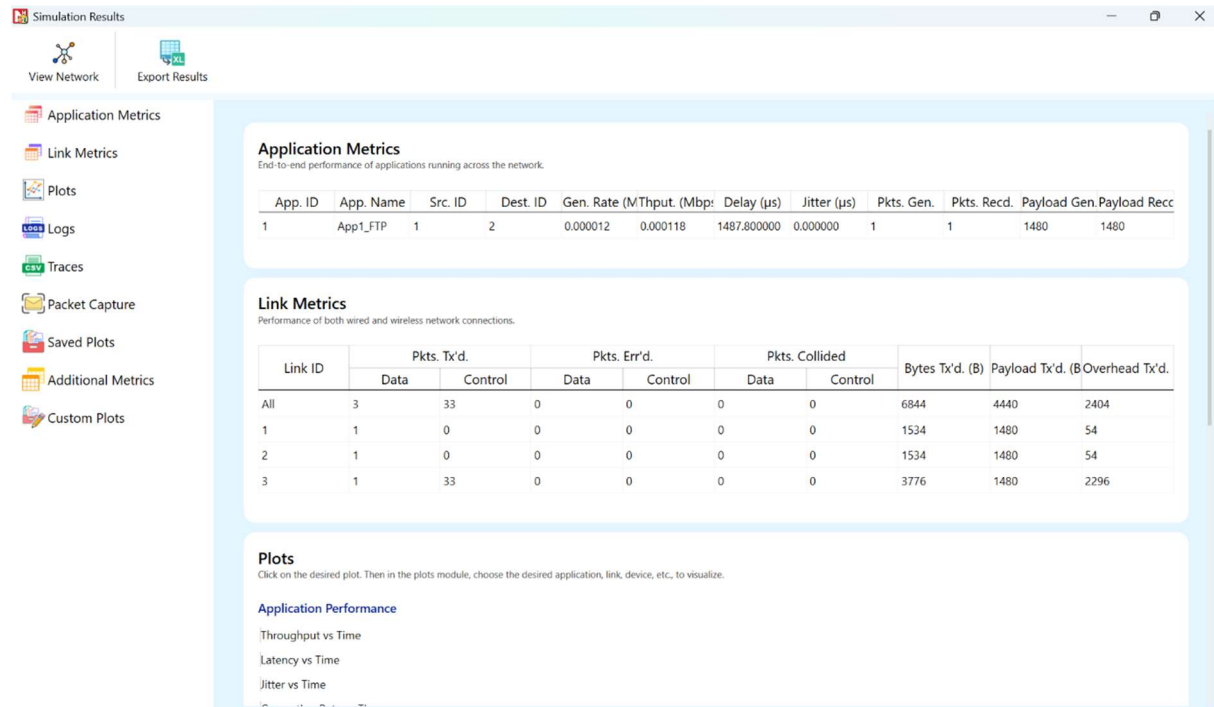
SETUP / PROCEDURE

1. Create a TCP network scenario with adjustable link error rate.
2. Set BER values (1%, 5%, 10%).
3. Observe retransmission counts, throughput, and transfer time.

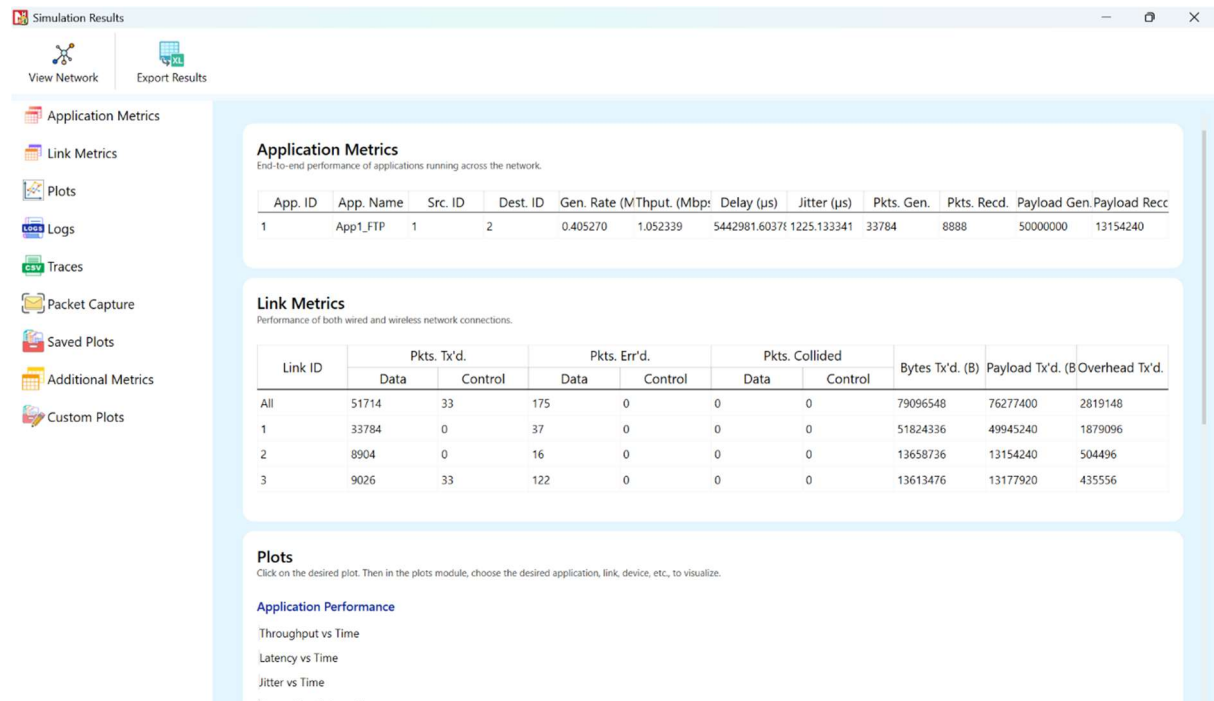
TOPOLOGY:



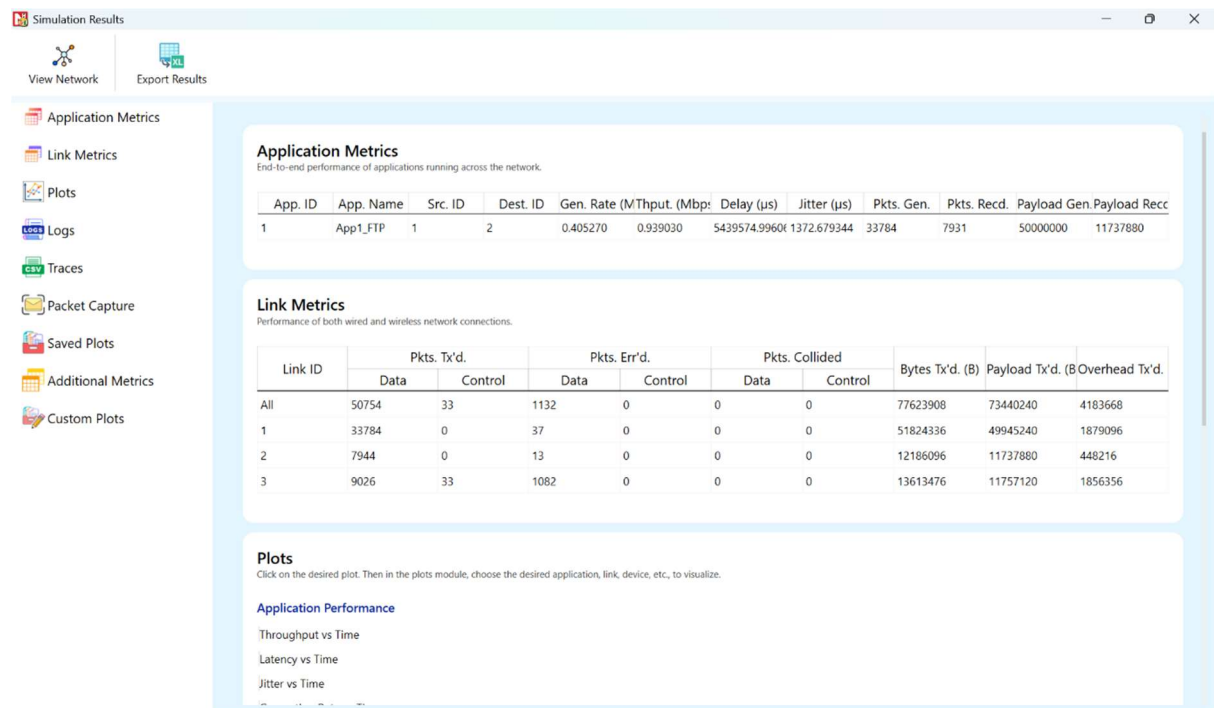
TESTCASE 1:



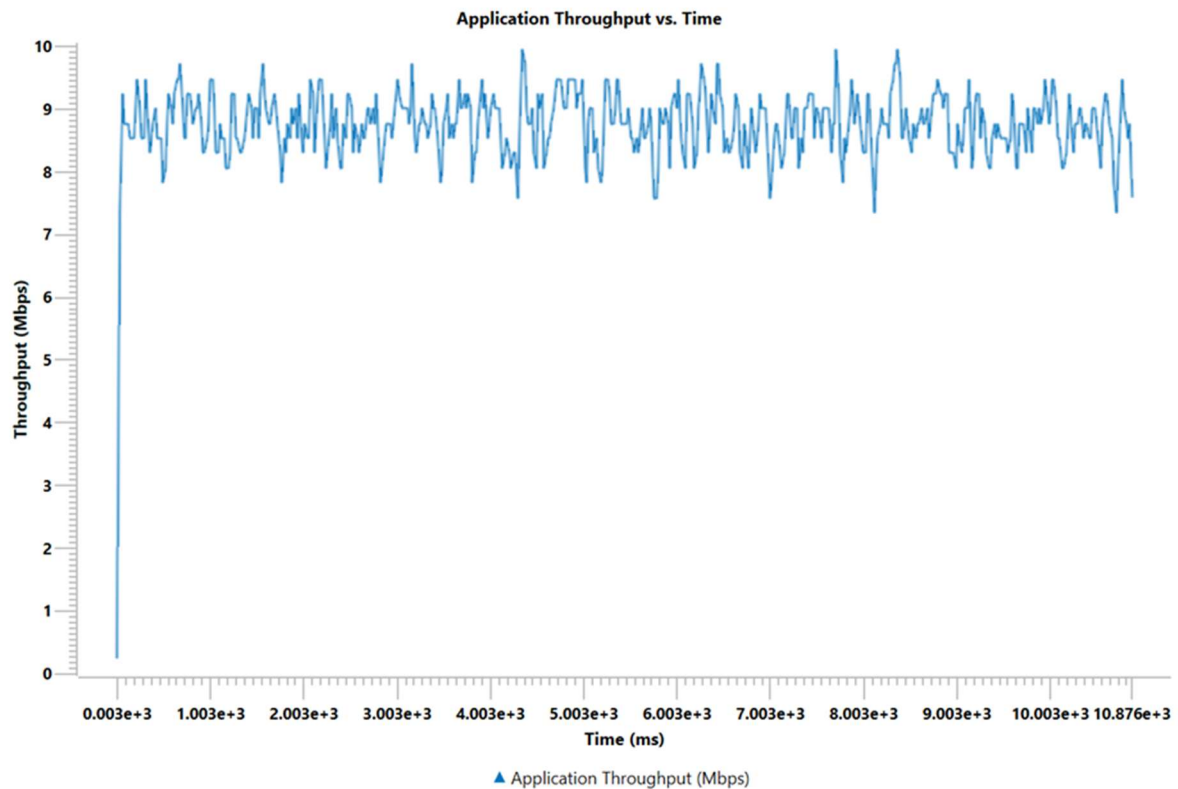
TESTCASE 2:

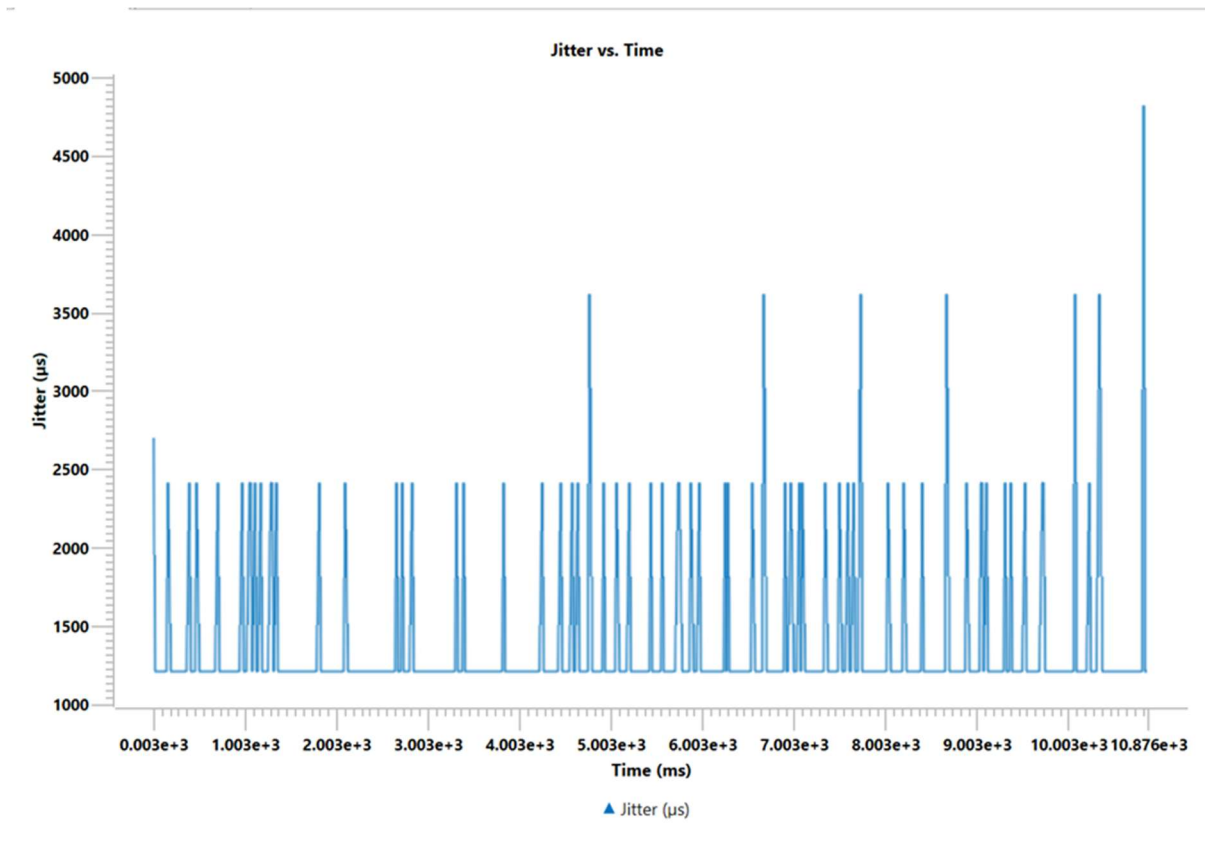
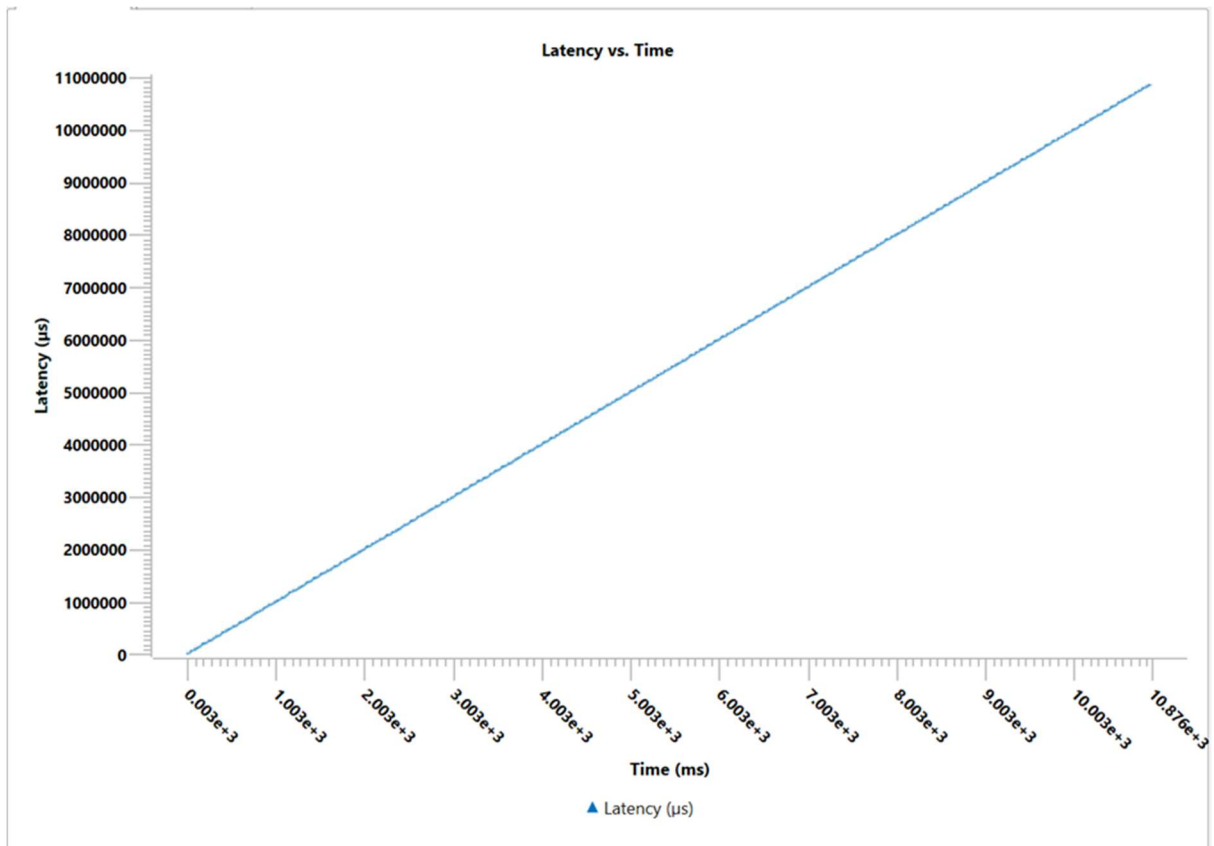


TESTCASE 3:



GRAPHS:





OBSERVATION

As packet loss increases, retransmissions rise, reducing throughput.

CONCLUSION

TCP maintains reliability at the cost of performance when packet loss occurs.

Experiment 4: UDP under Packet Loss

AIM

To study UDP performance in networks experiencing packet loss.

INTRODUCTION

Since UDP lacks retransmission and acknowledgment, packet loss directly reduces data accuracy and quality.

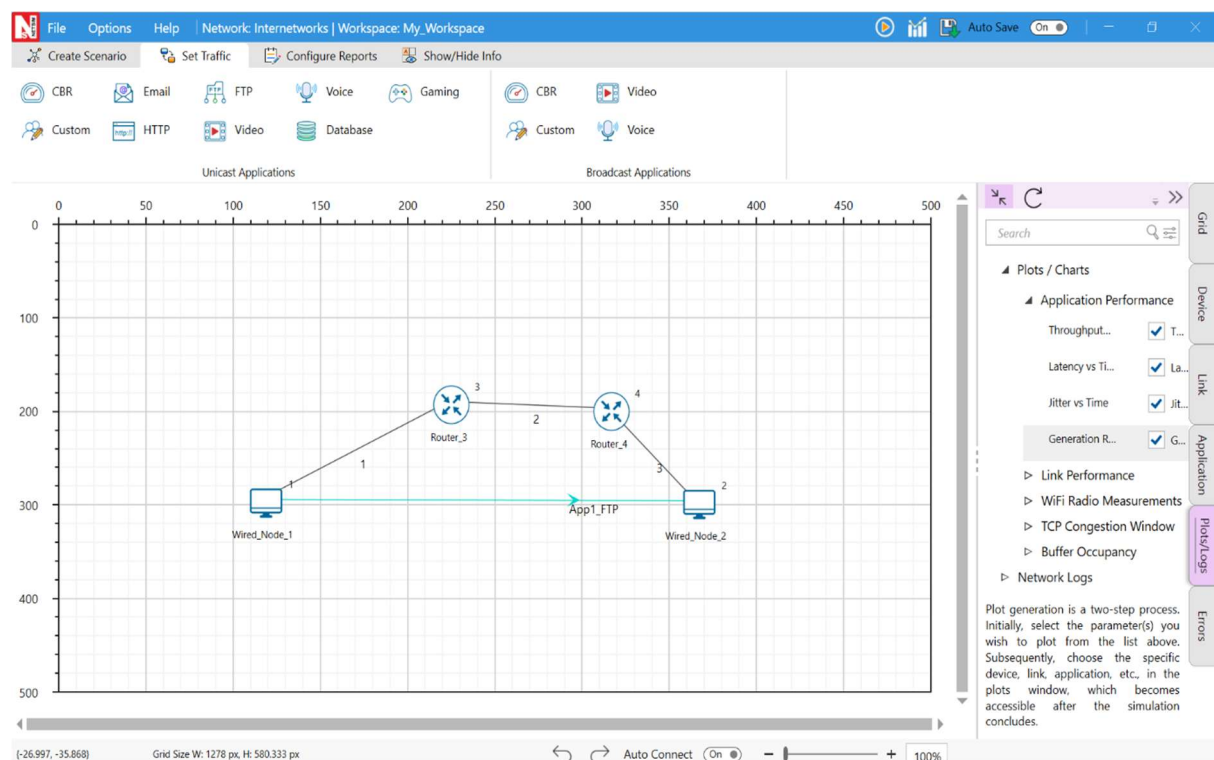
THEORY

UDP sends packets without error recovery. Any lost packet is permanently dropped, which affects real-time applications such as video or voice streaming.

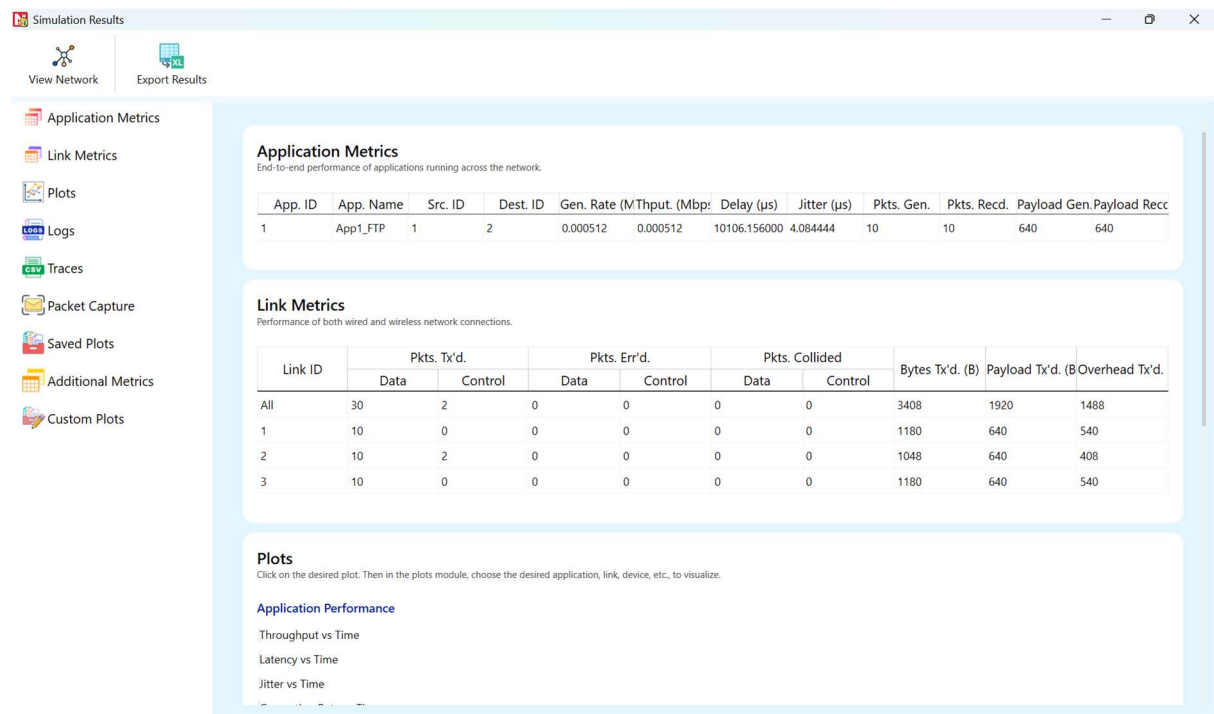
SETUP / PROCEDURE

1. Set up a UDP network and introduce different BER levels.
2. Record packet loss rate, delay, and jitter.
3. Compare results under increasing loss.

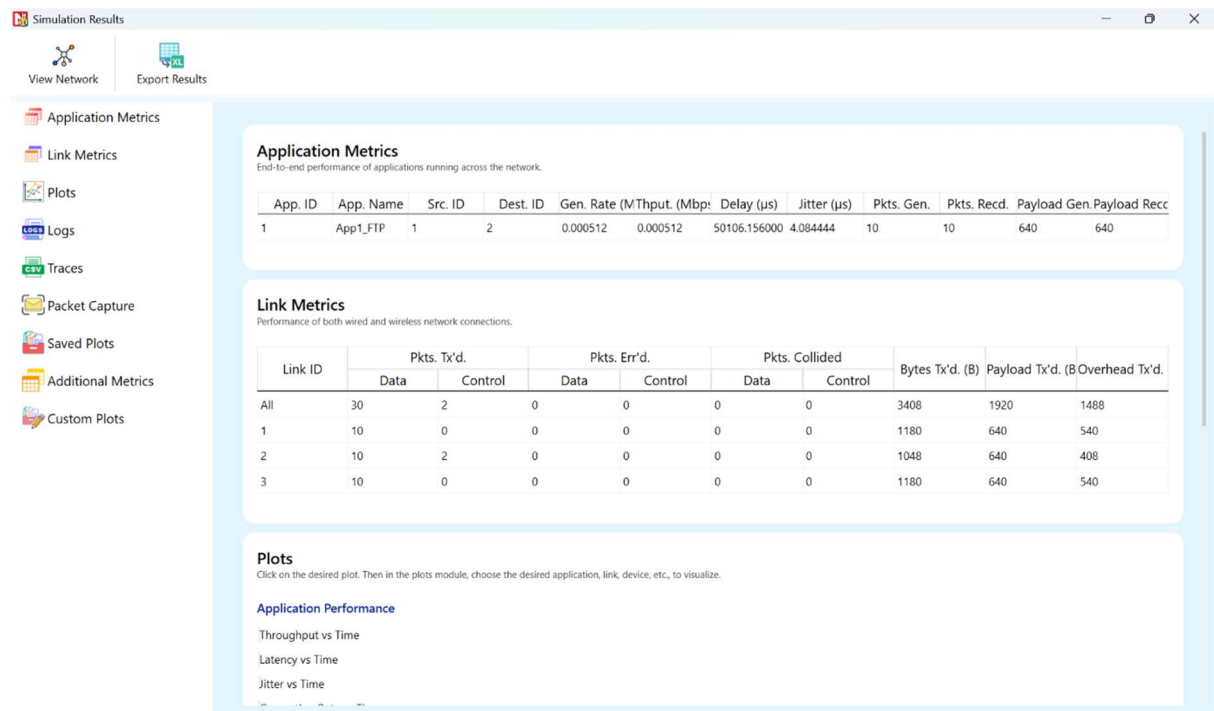
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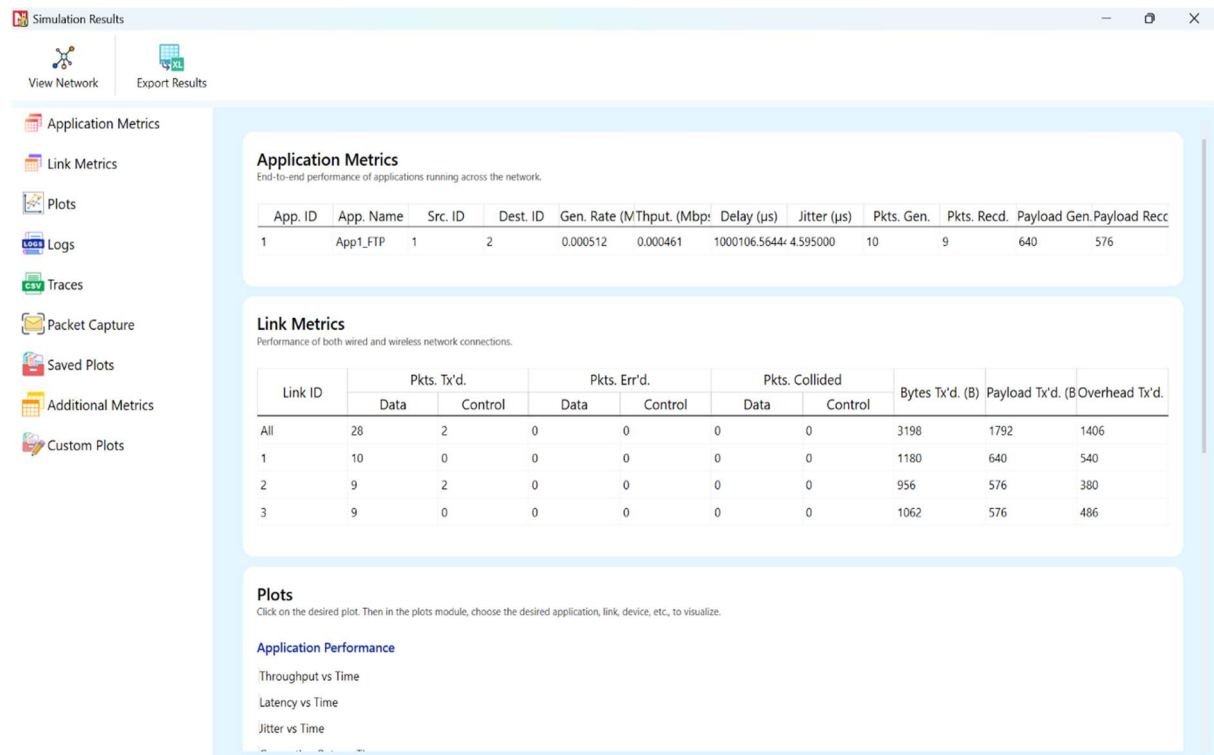
TESTCASE 1:



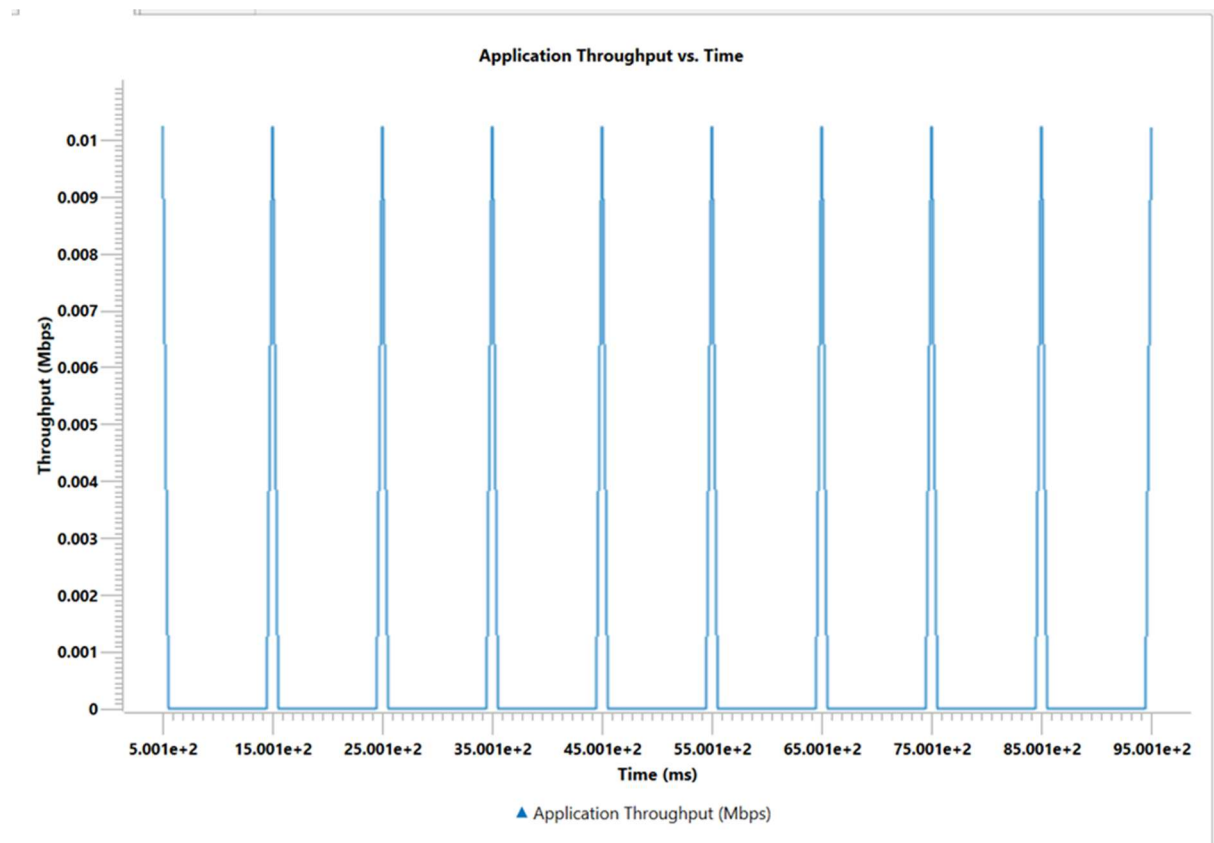
TESTCASE 2:

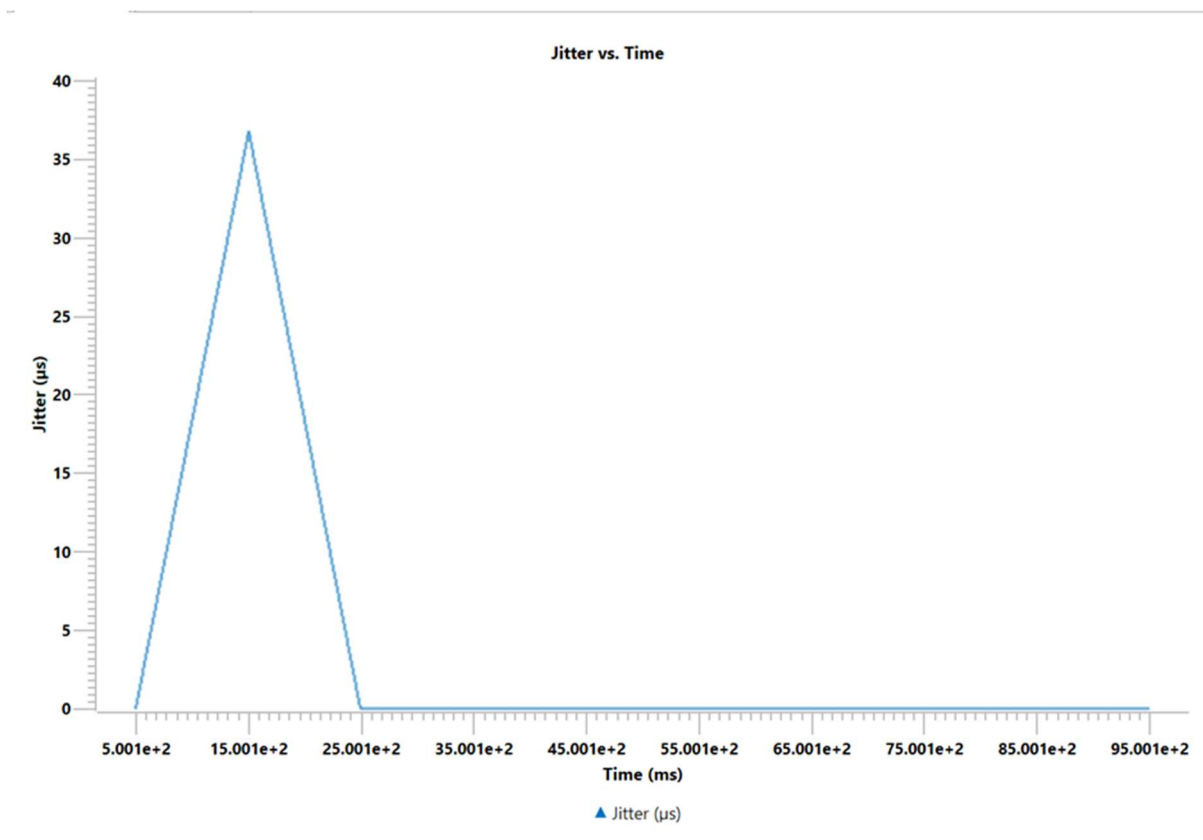
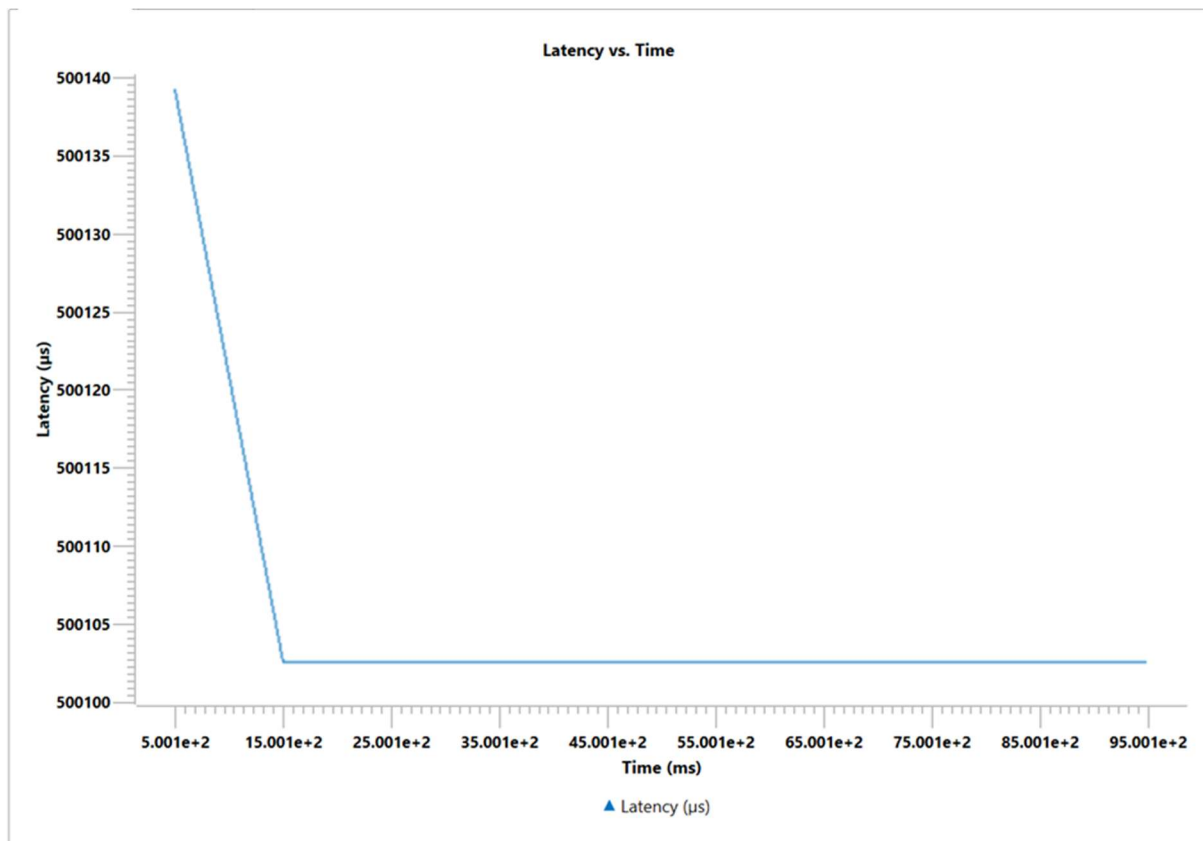


TESTCASE 3:



GRAPHS:





OBSERVATION

Packet loss increases linearly with BER, with no retransmission recovery.

CONCLUSION

UDP performance degrades under packet loss, showing why it is best for tolerant real-time traffic.

Experiment 5: FTP Transfer Performance

AIM

To evaluate the performance of FTP (File Transfer Protocol) under various network conditions.

INTRODUCTION

FTP operates on top of TCP and provides reliable file transfer services. Its performance depends on the underlying TCP behaviour.

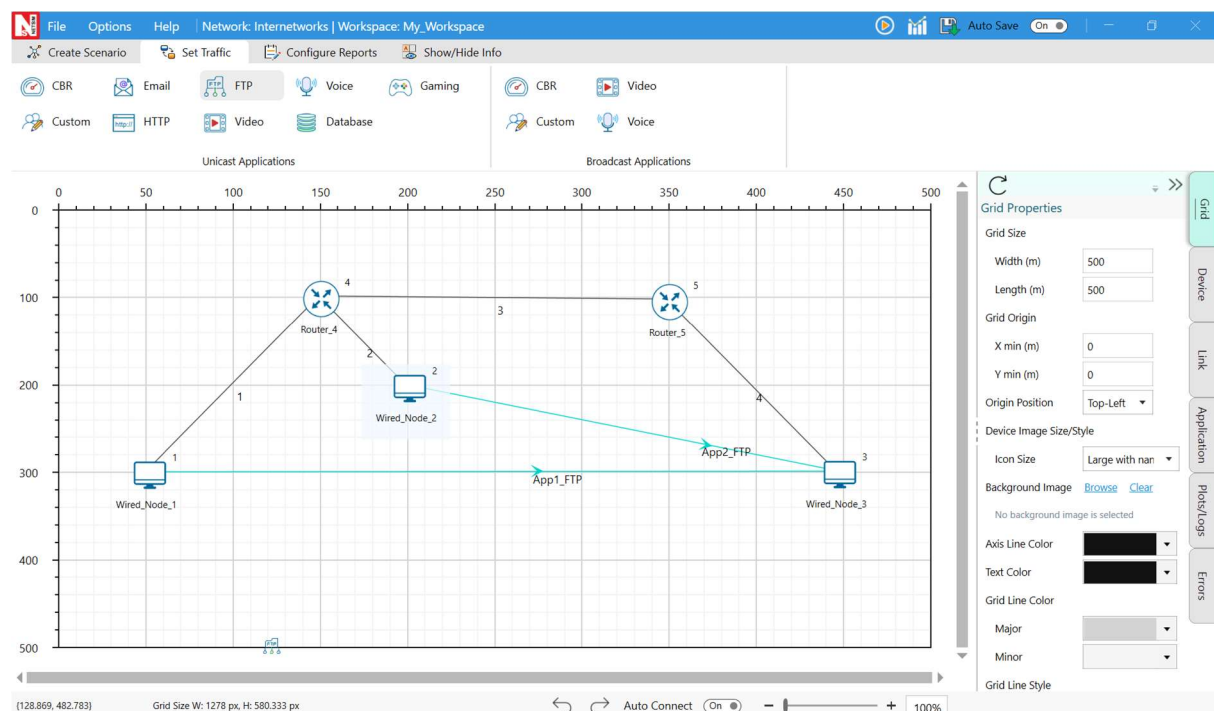
THEORY

Since FTP relies on TCP, factors like congestion, delay, and loss affect throughput. When congestion increases, FTP transfers become slower due to TCP's congestion avoidance mechanism.

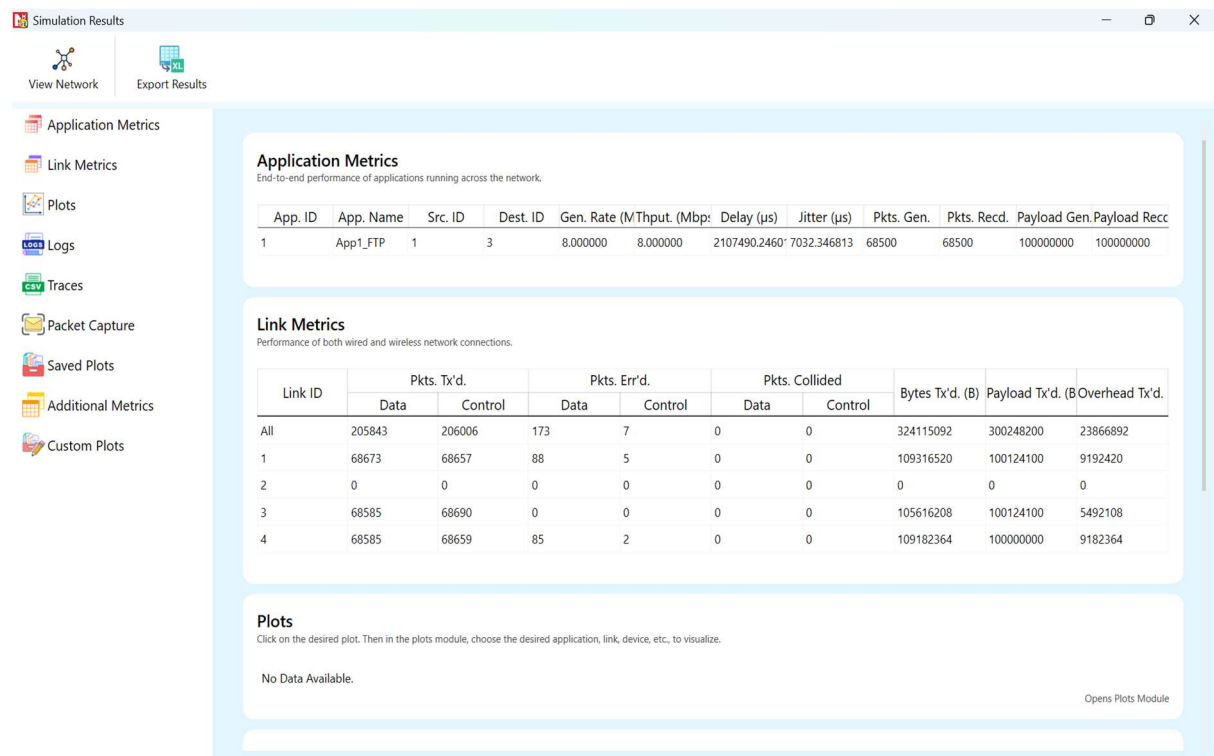
SETUP / PROCEDURE

1. Create an FTP client-server setup in NetSim Lite.
2. Vary bandwidth and background traffic.
3. Measure FTP throughput, delay, and transfer time.

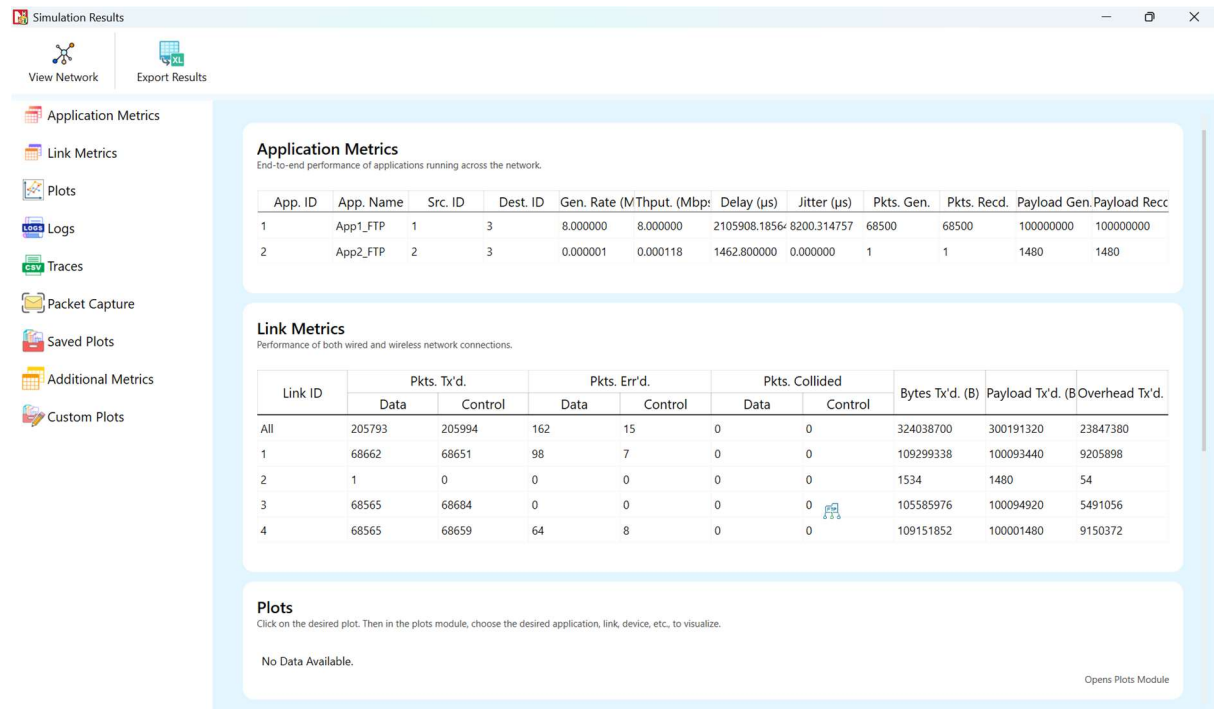
TOPOLOGY



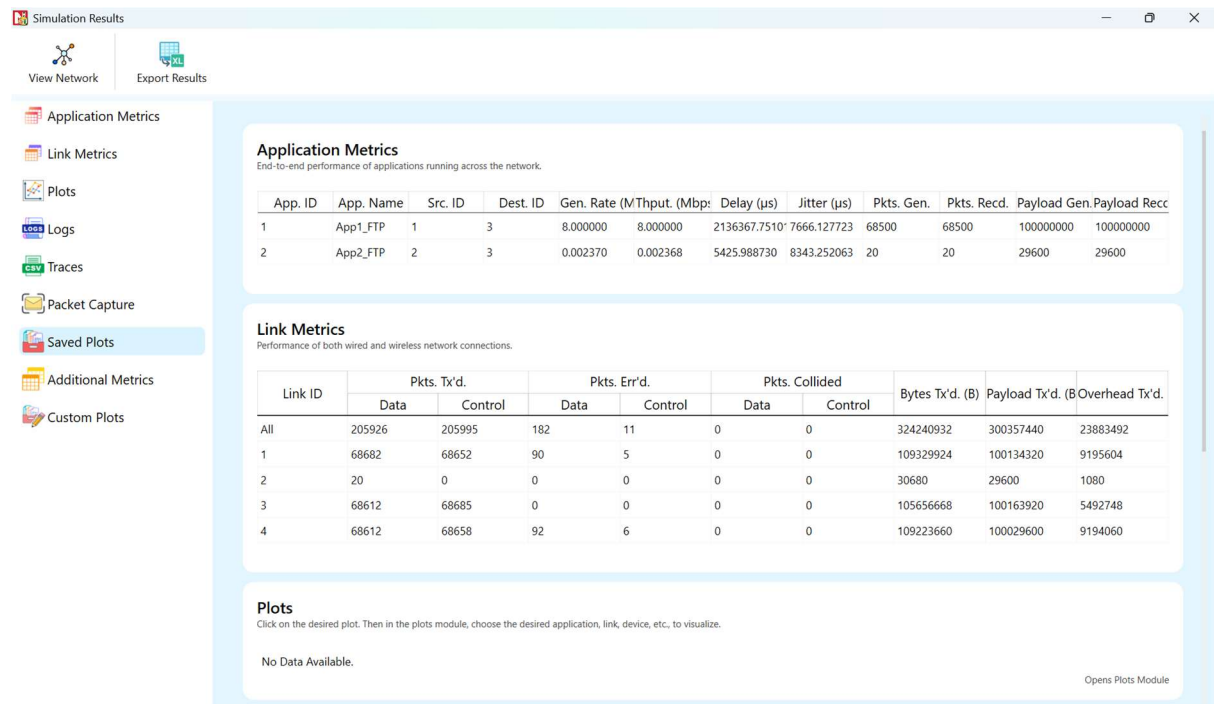
TESTCASE 1:



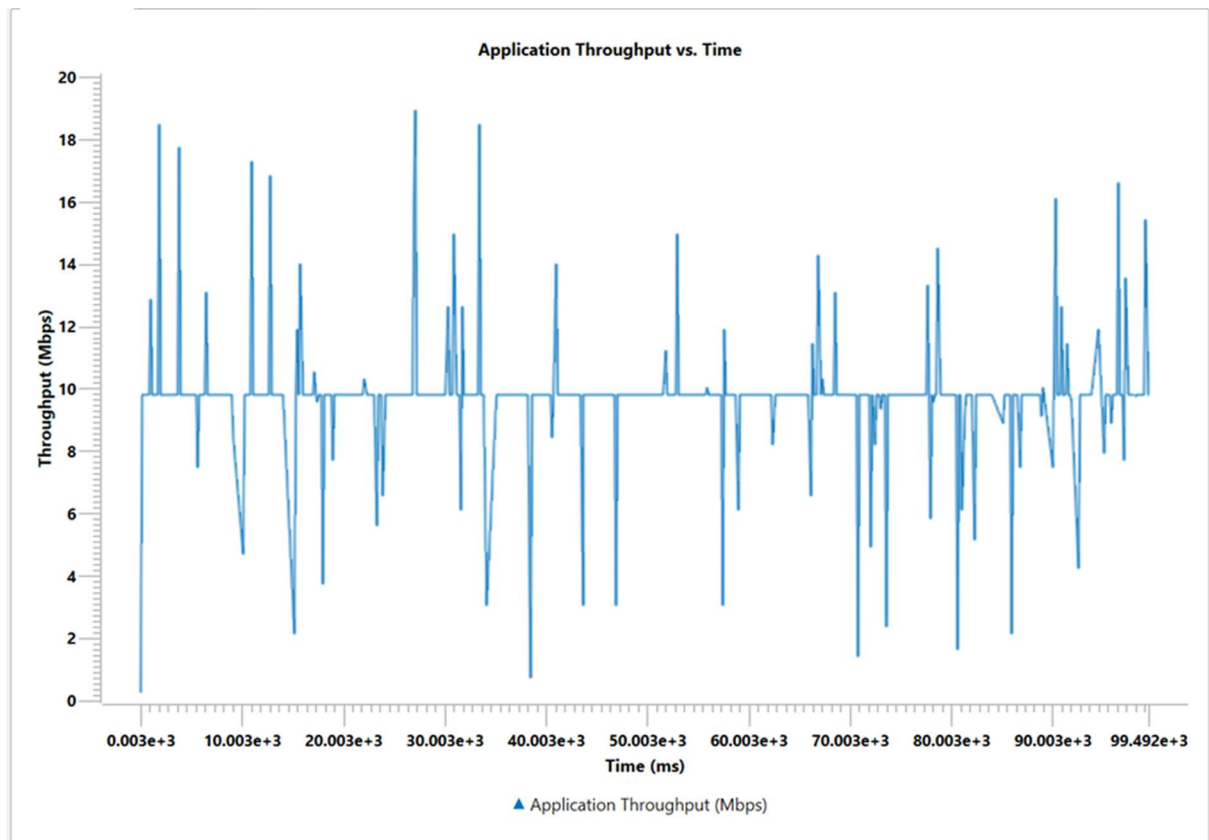
TESTCASE 2:

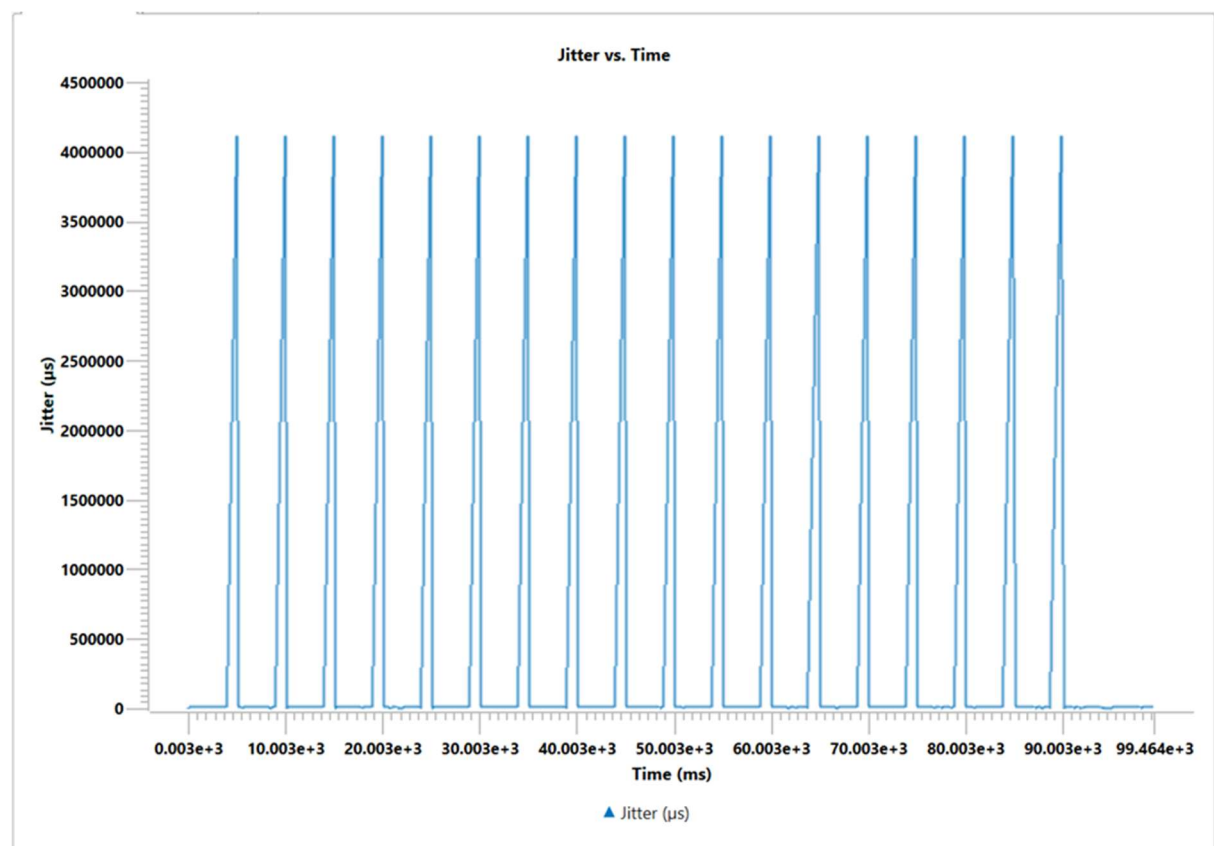
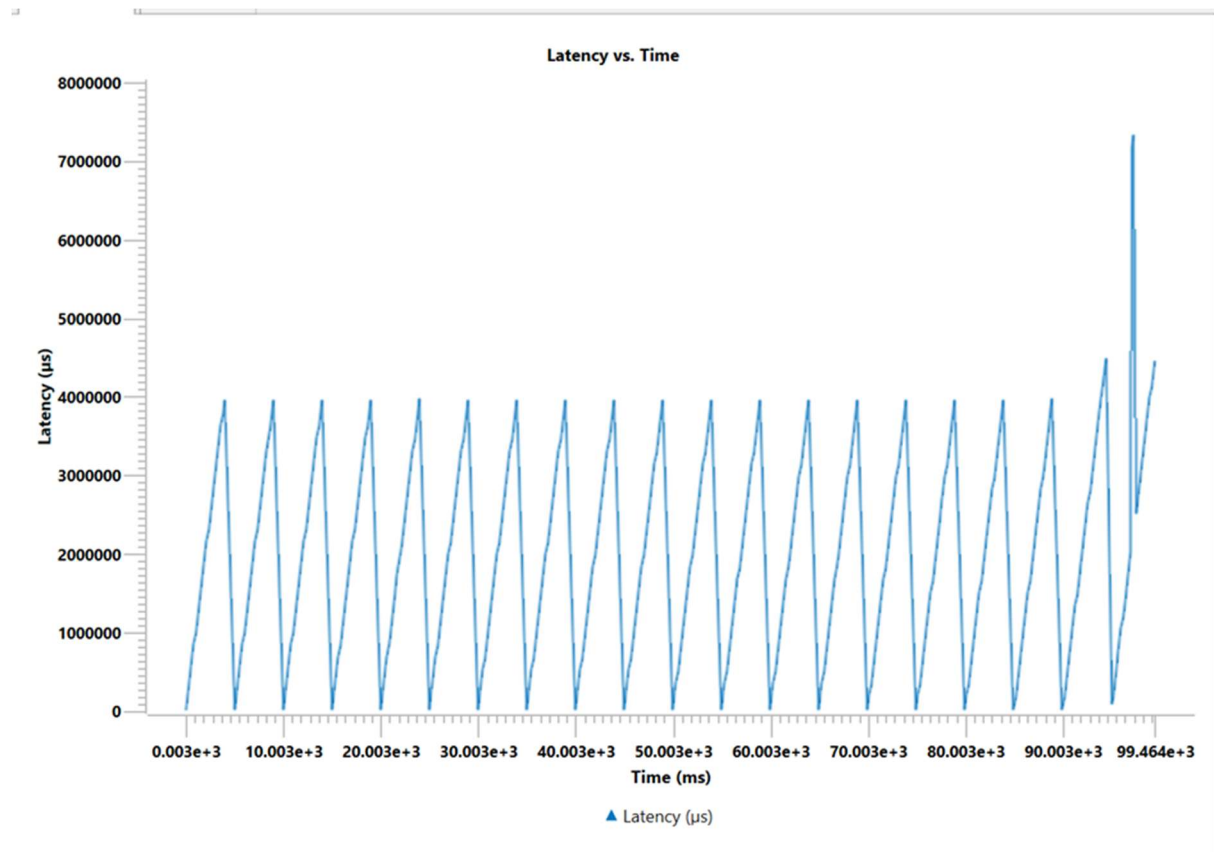


TESTCASE 3:



GRAPHS :





OBSERVATION

FTP throughput decreases during congestion or high delay.

CONCLUSION

FTP achieves reliable delivery, but performance declines when TCP experiences congestion.

Experiment 6: HTTP Web Browsing Performance

AIM

To analyse the performance of HTTP-based web browsing in different network conditions.

INTRODUCTION

HTTP runs on TCP and retrieves web resources such as HTML, images, and scripts. Its performance depends on network latency and number of parallel connections.

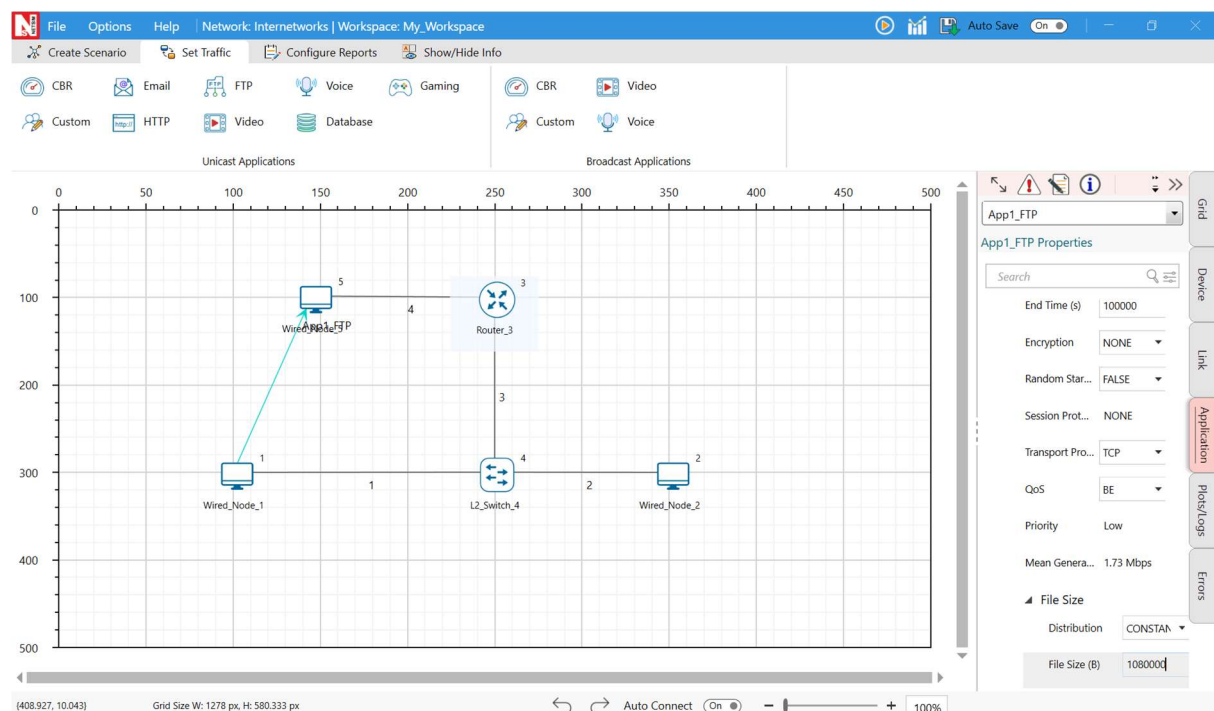
THEORY

Each HTTP request involves a TCP handshake and data transfer. Multiple connections (HTTP/1.1 or 2.0) can improve performance but increase overhead.

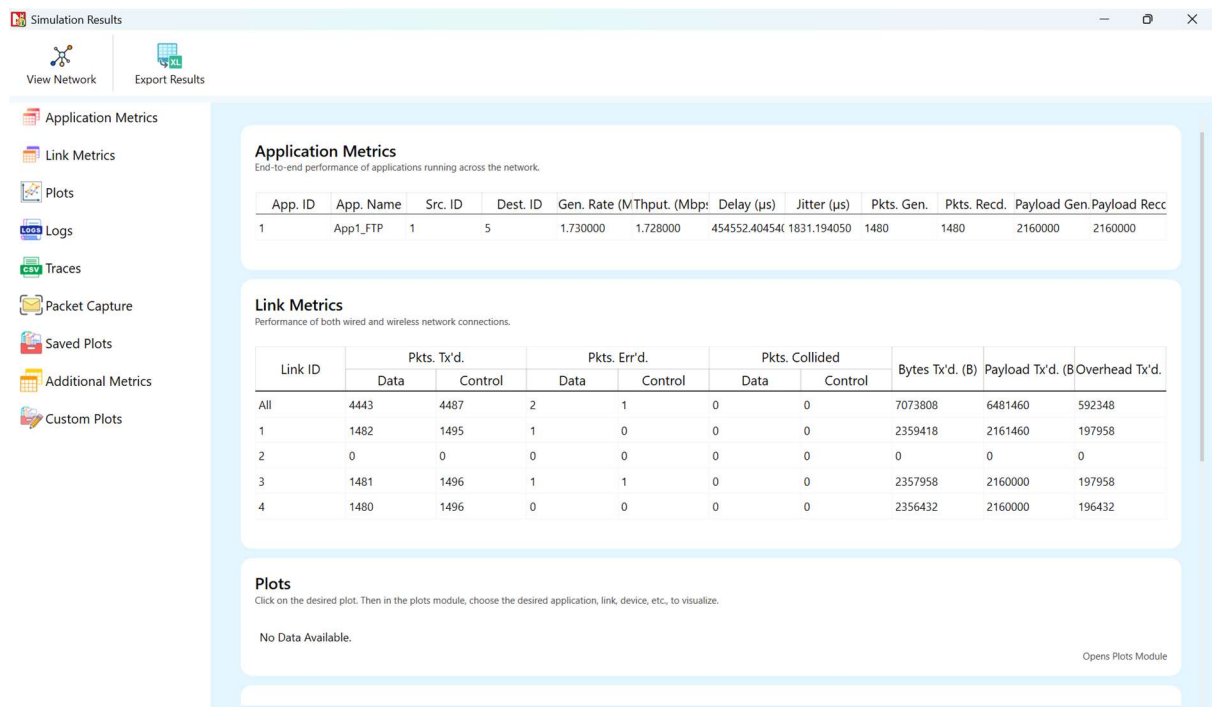
SETUP / PROCEDURE

1. Configure an HTTP client-server scenario in NetSim.
2. Simulate one large transfer vs. multiple small transfers.
3. Measure total transfer time and delay.

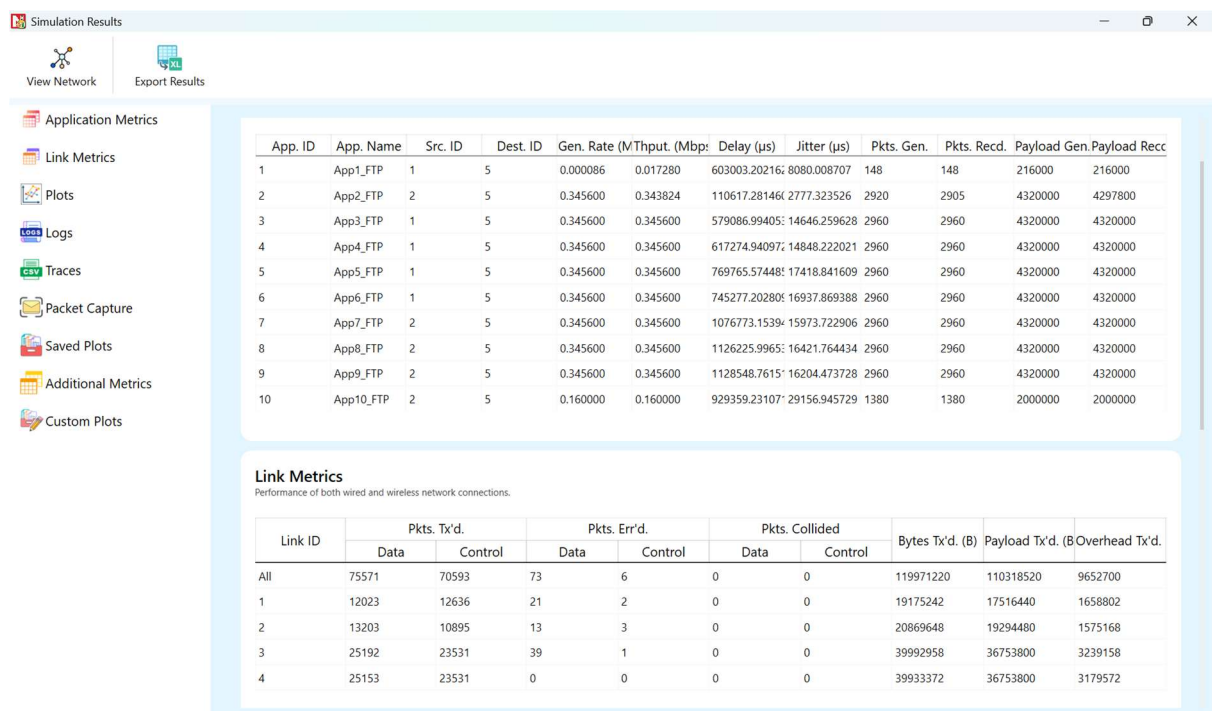
TOPOLOGY



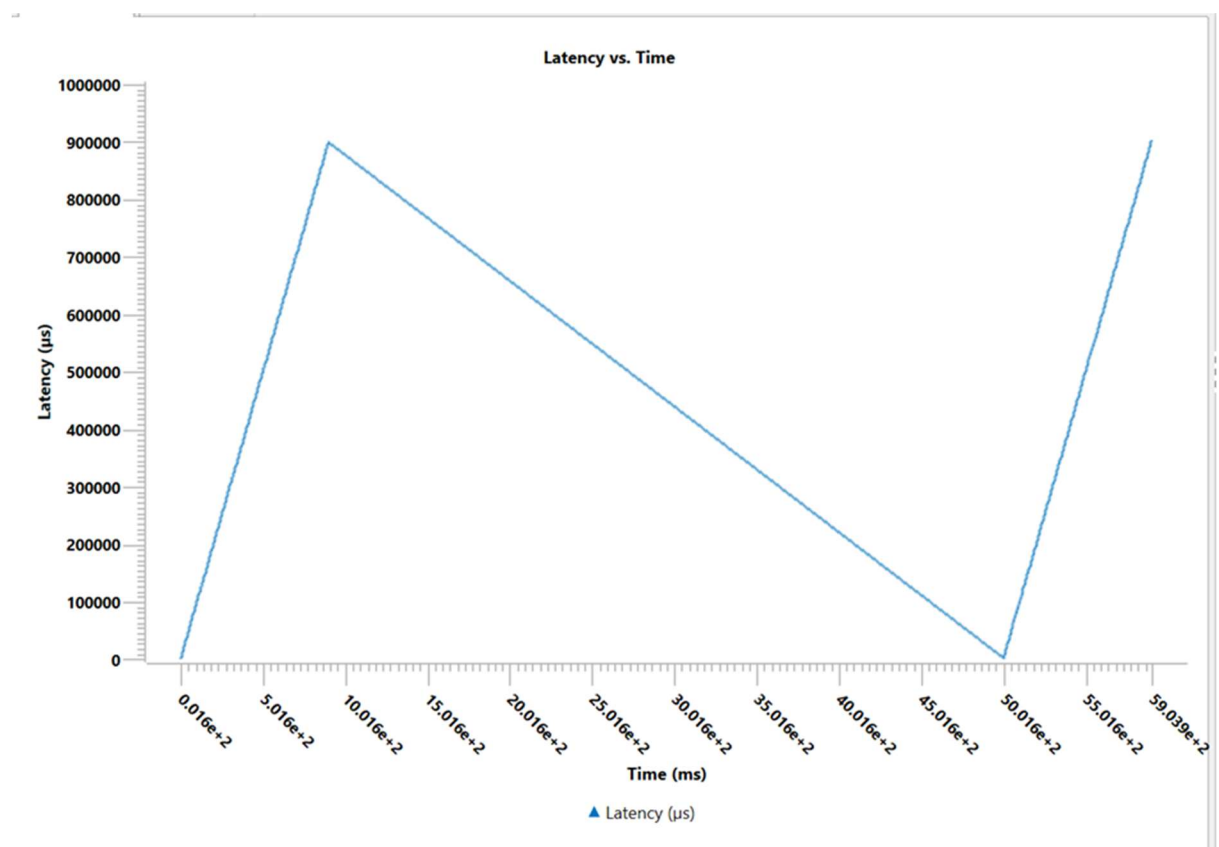
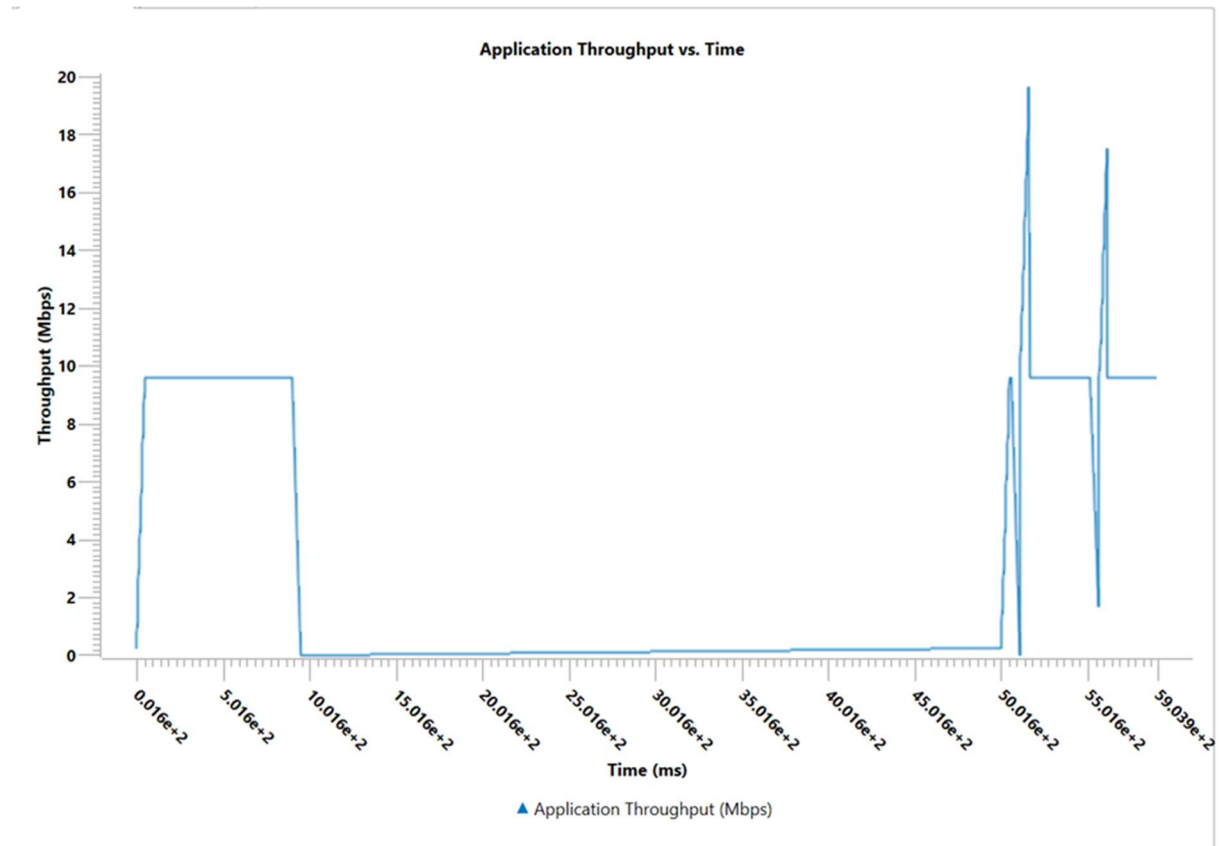
TESTCASE 1:

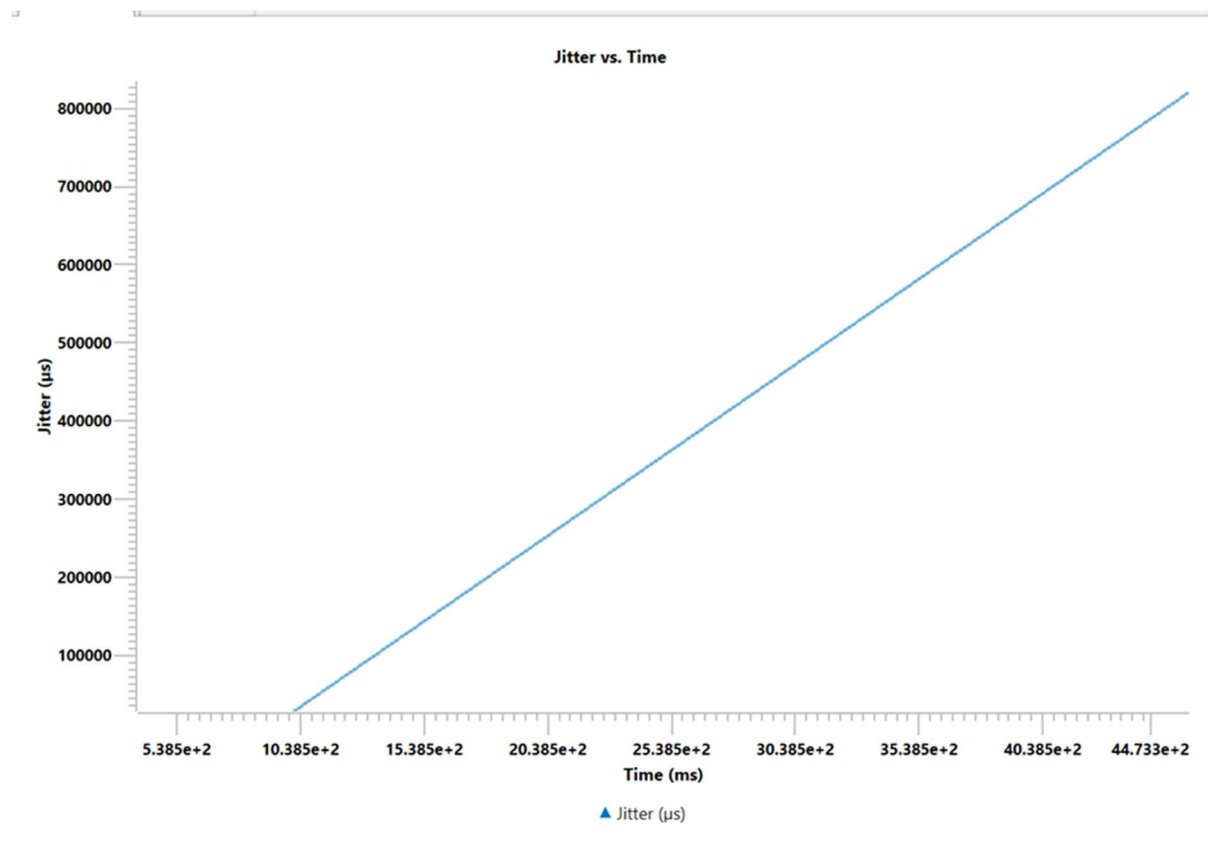


TESTCASE 2:



GRAPHS:





OBSERVATION

Single large transfers are efficient, while multiple small connections cause extra setup delay.

CONCLUSION

HTTP performance improves with fewer connections and lower latency.

Experiment 7: VoIP over TCP vs UDP

AIM

To compare VoIP performance using TCP and UDP as transport protocols.

INTRODUCTION

VoIP applications require low delay and jitter. UDP is preferred because it sends data continuously without retransmissions, while TCP introduces latency through acknowledgments.

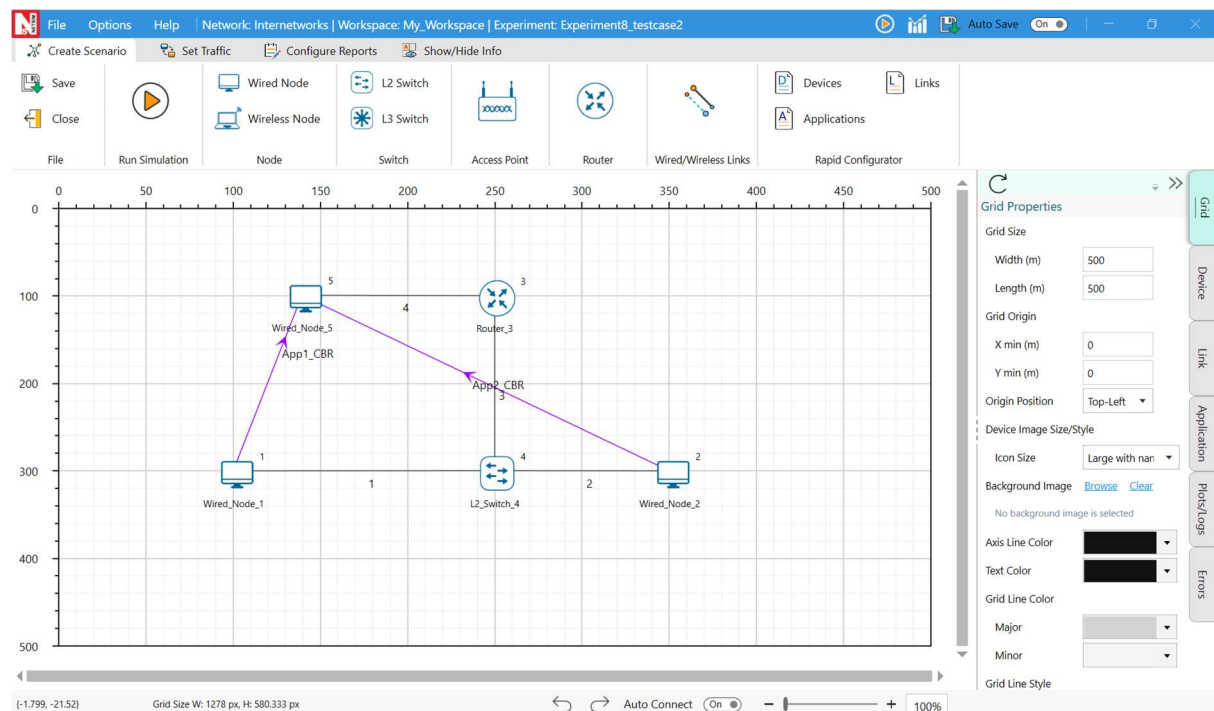
THEORY

VoIP streams are real-time and sensitive to delay. Retransmissions in TCP disrupt audio flow, whereas UDP allows minor packet loss without breaking continuity.

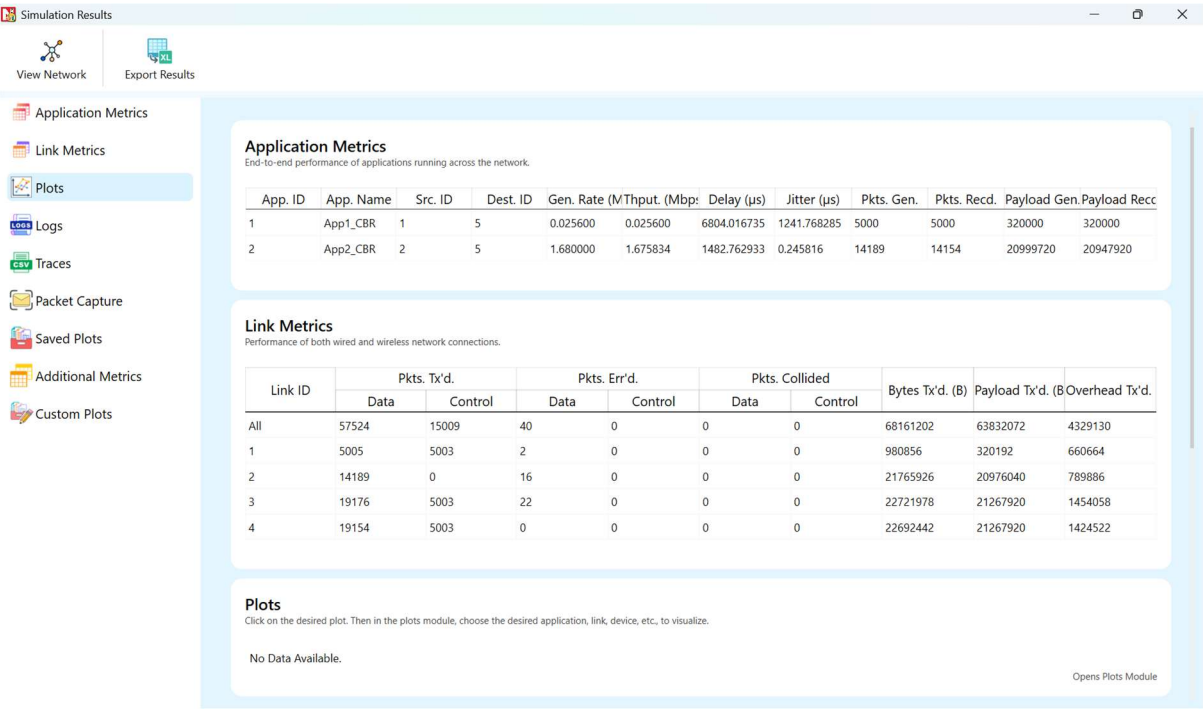
SETUP / PROCEDURE

1. Create a VoIP session using TCP and UDP.
2. Measure delay, jitter, and packet loss.
3. Compare the two protocols.

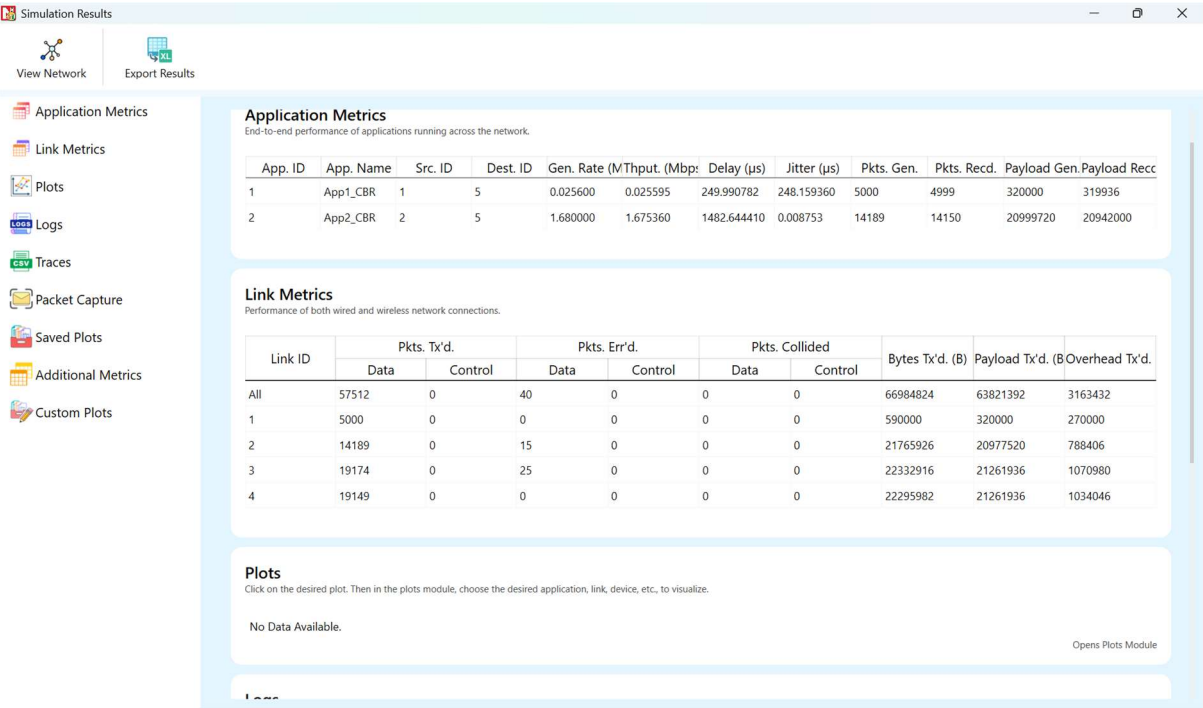
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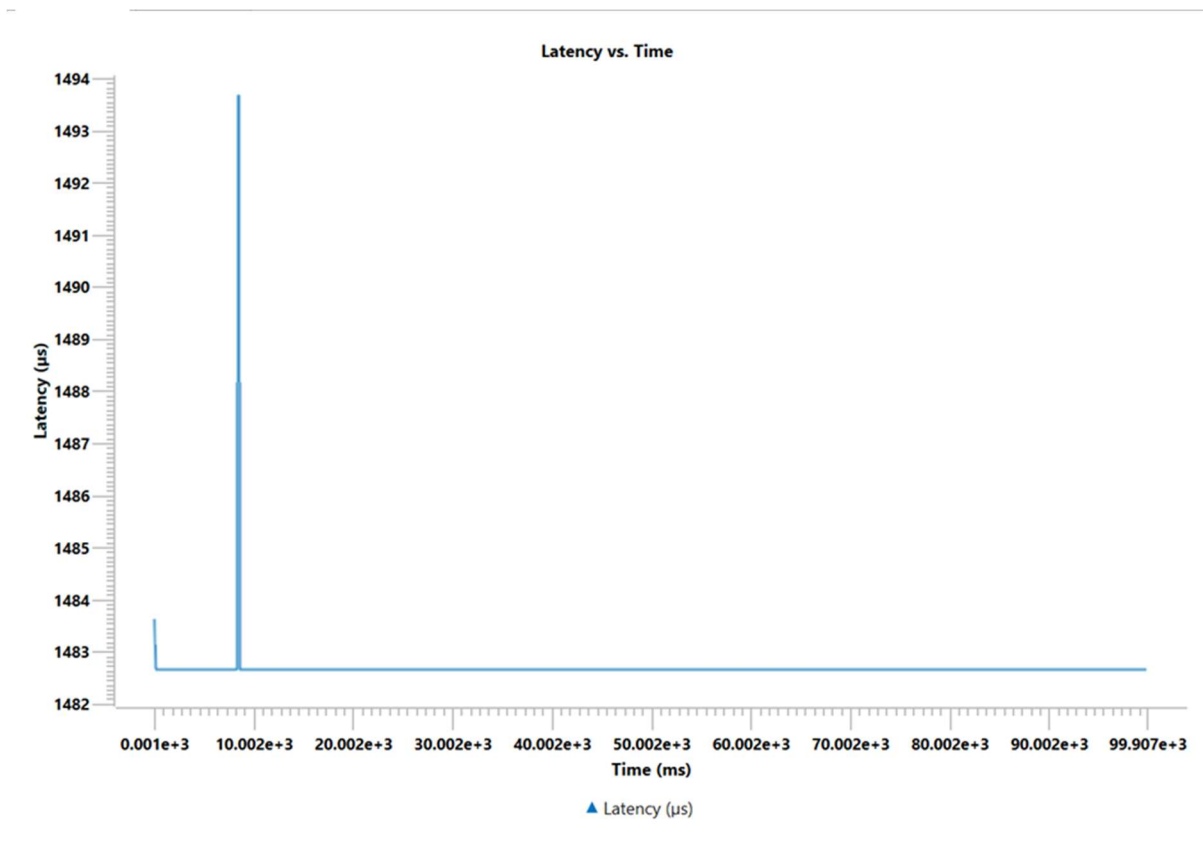
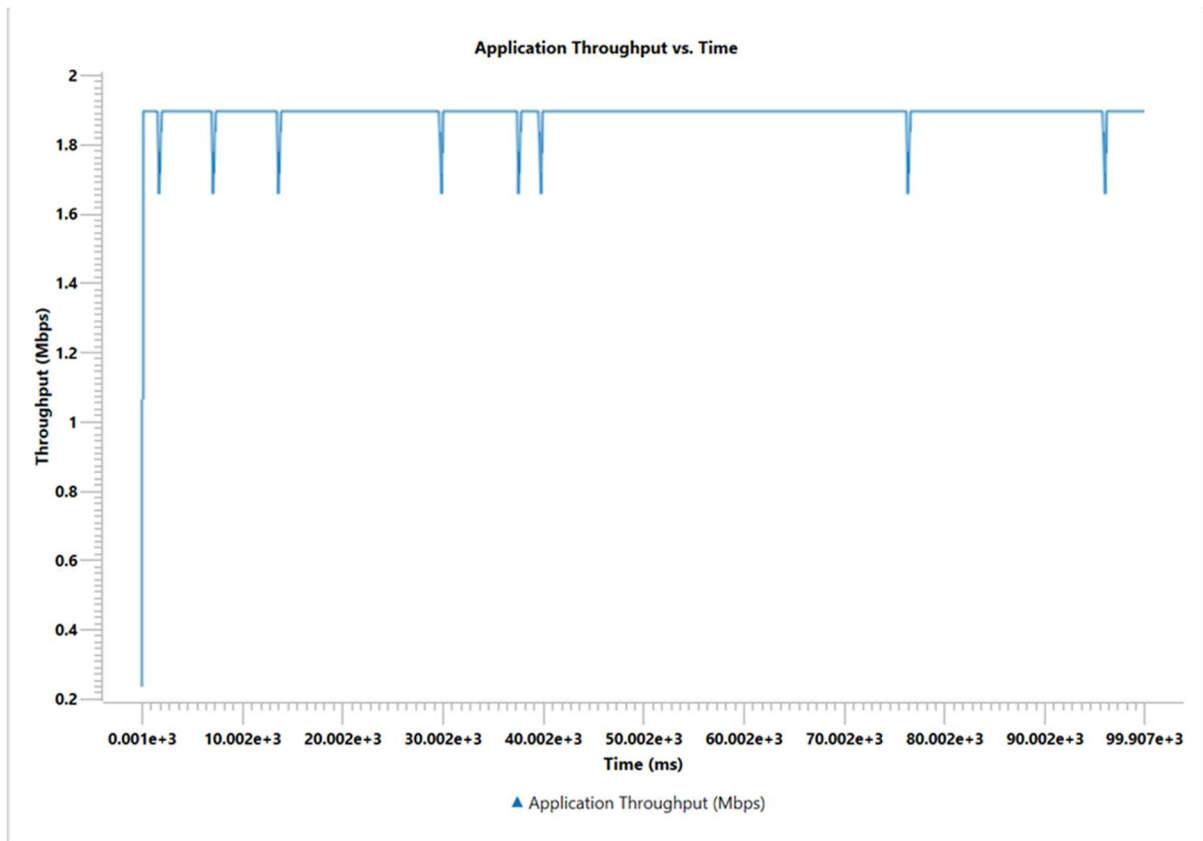
TESTCASE 1:

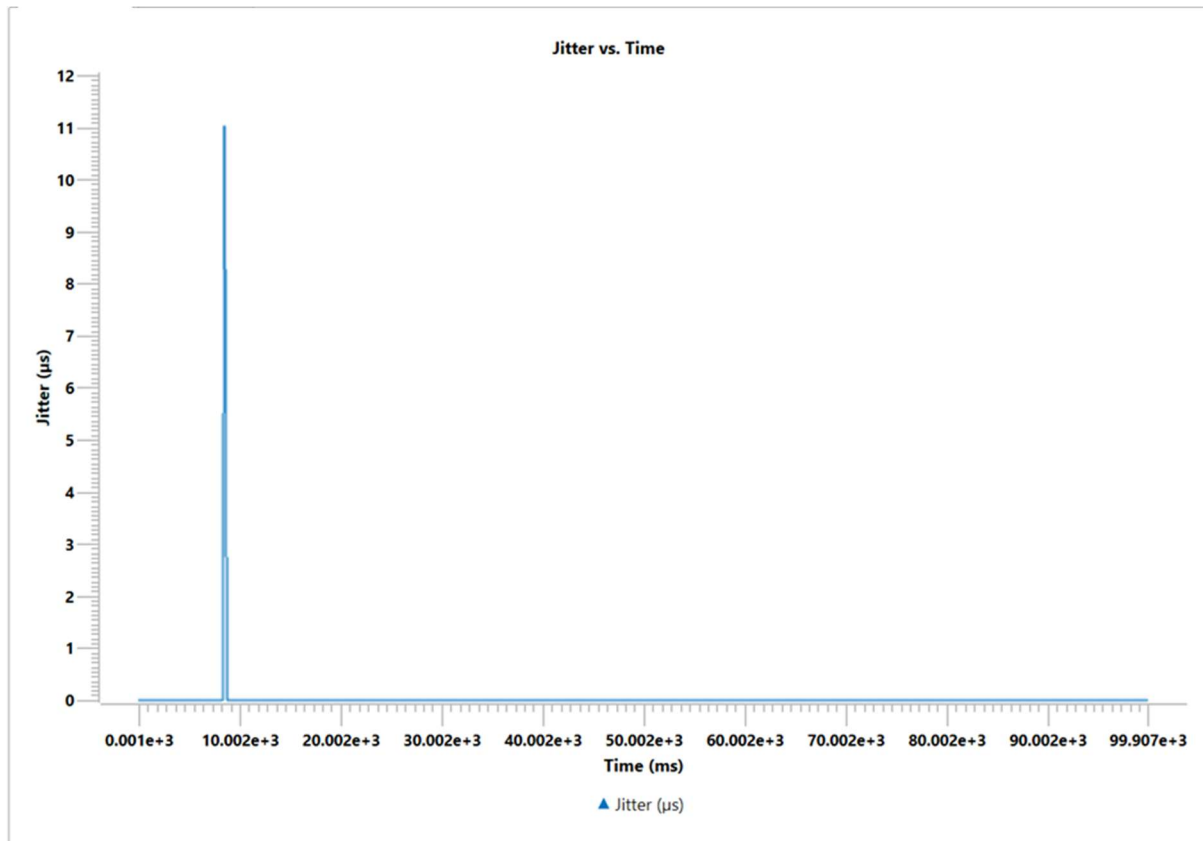


TESTCASE 2:



GRAPHS :





OBSERVATION

UDP shows lower delay and jitter, while TCP introduces retransmission delays.

CONCLUSION

UDP is more suitable for VoIP due to lower latency, despite minor packet losses.

Experiment 8: Network Congestion Test

AIM

To observe how multiple competing protocols (TCP and UDP) share bandwidth and experience congestion.

INTRODUCTION

When several flows share a link, congestion may occur if offered load exceeds capacity. TCP reduces its sending rate, while UDP continues sending, possibly starving TCP flows.

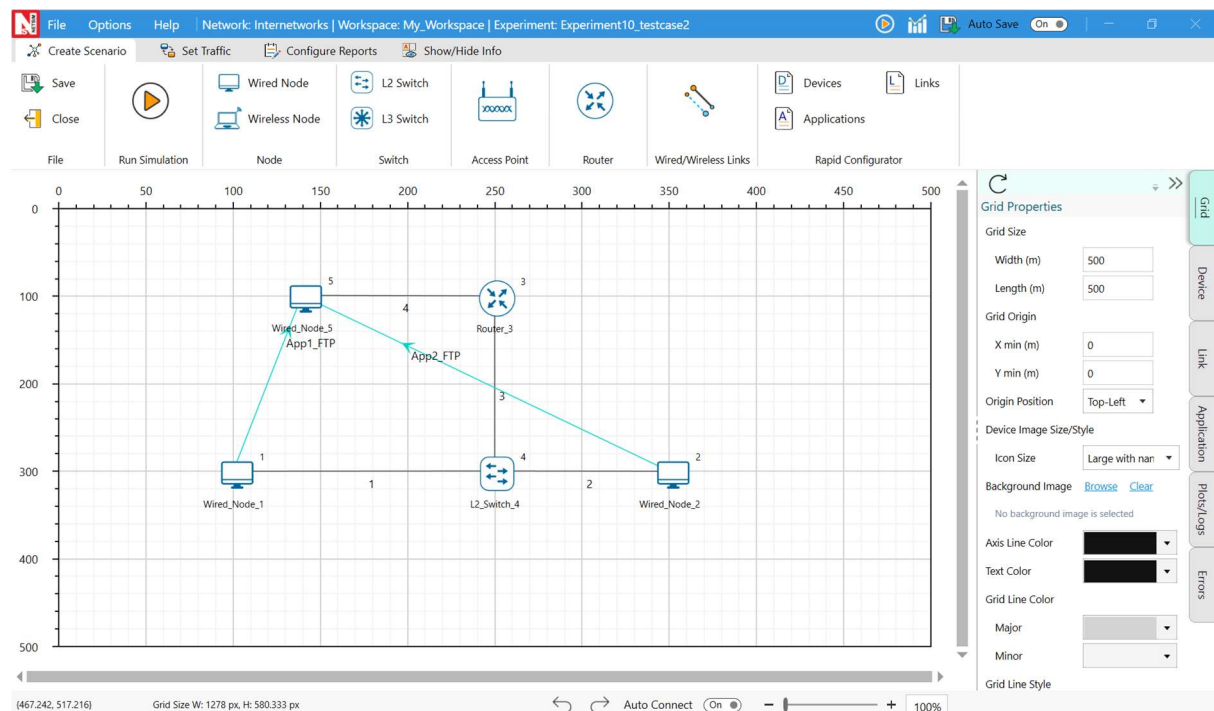
THEORY

TCP's congestion control adjusts window size on detecting packet loss. UDP, being uncontrolled, can dominate bandwidth, causing unfair throughput distribution.

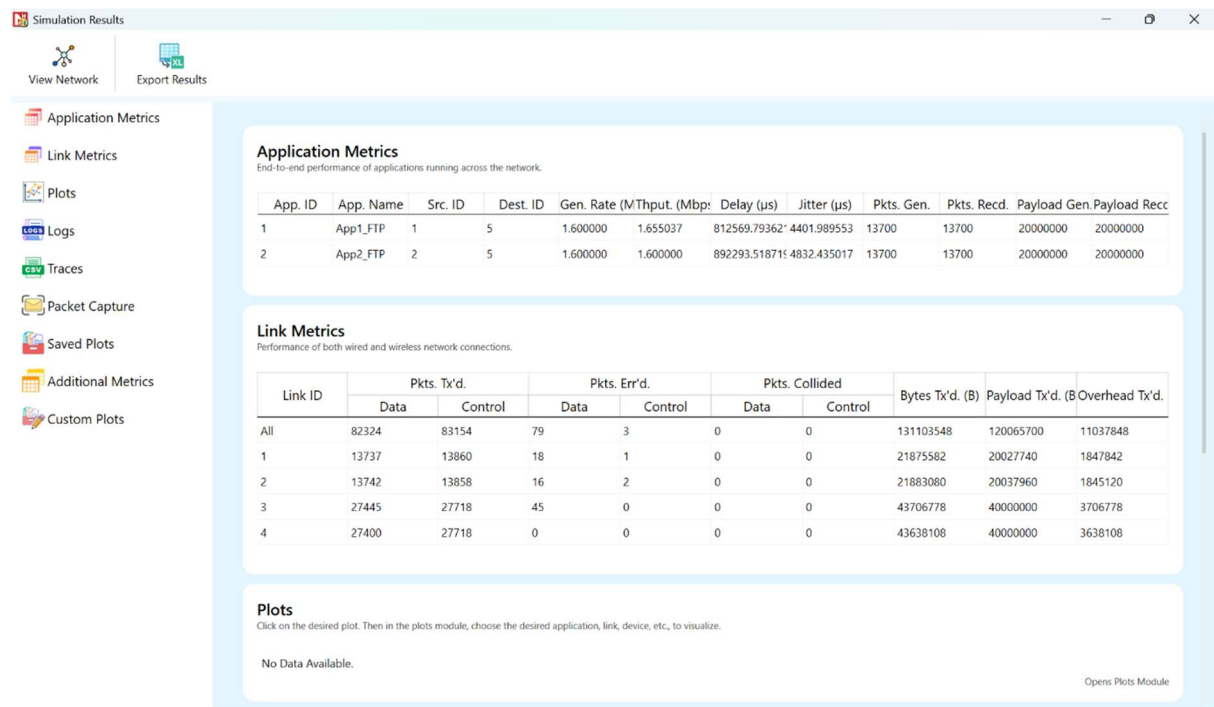
SETUP / PROCEDURE

1. Create a mixed traffic scenario (TCP + UDP).
2. Vary UDP background traffic rate.
3. Record throughput, fairness, and packet drops.

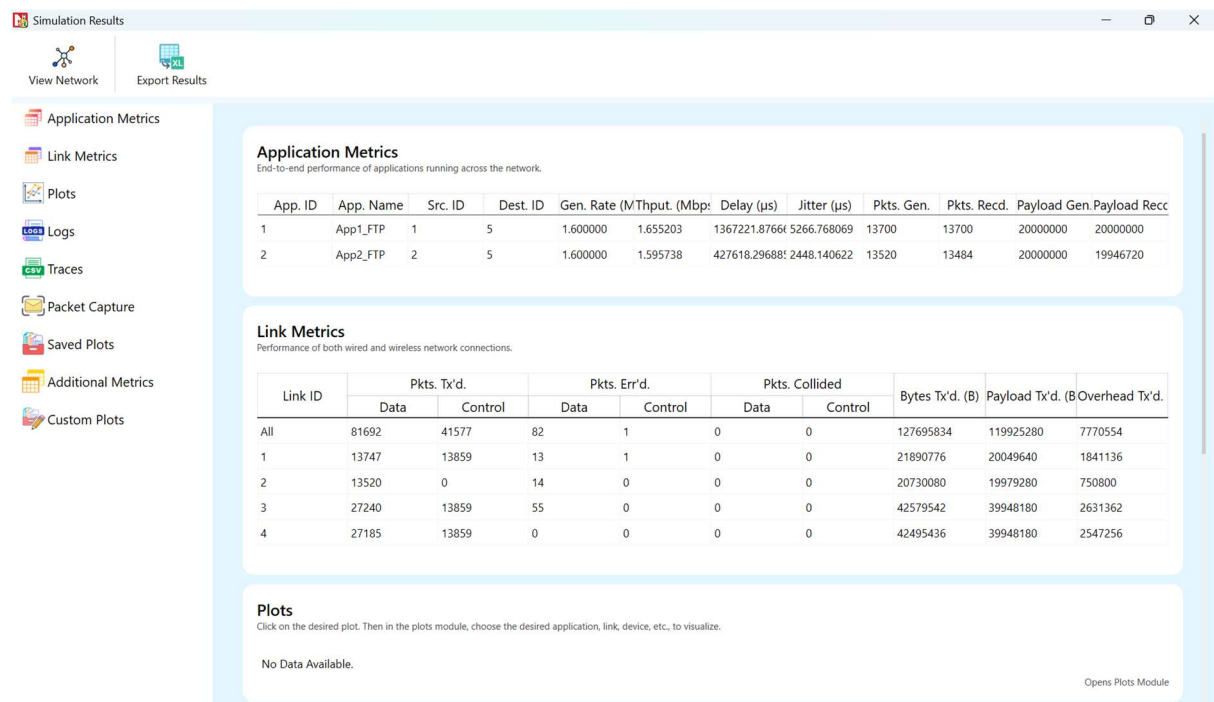
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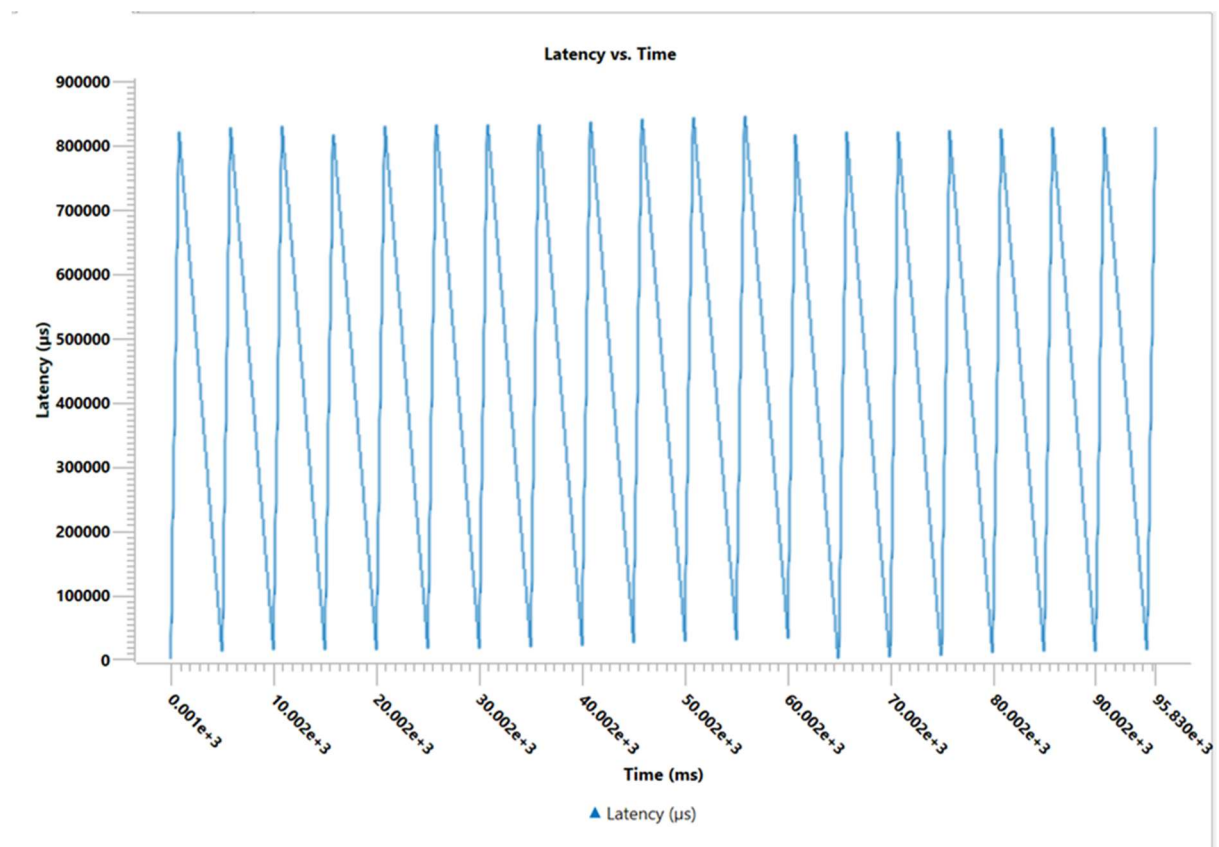
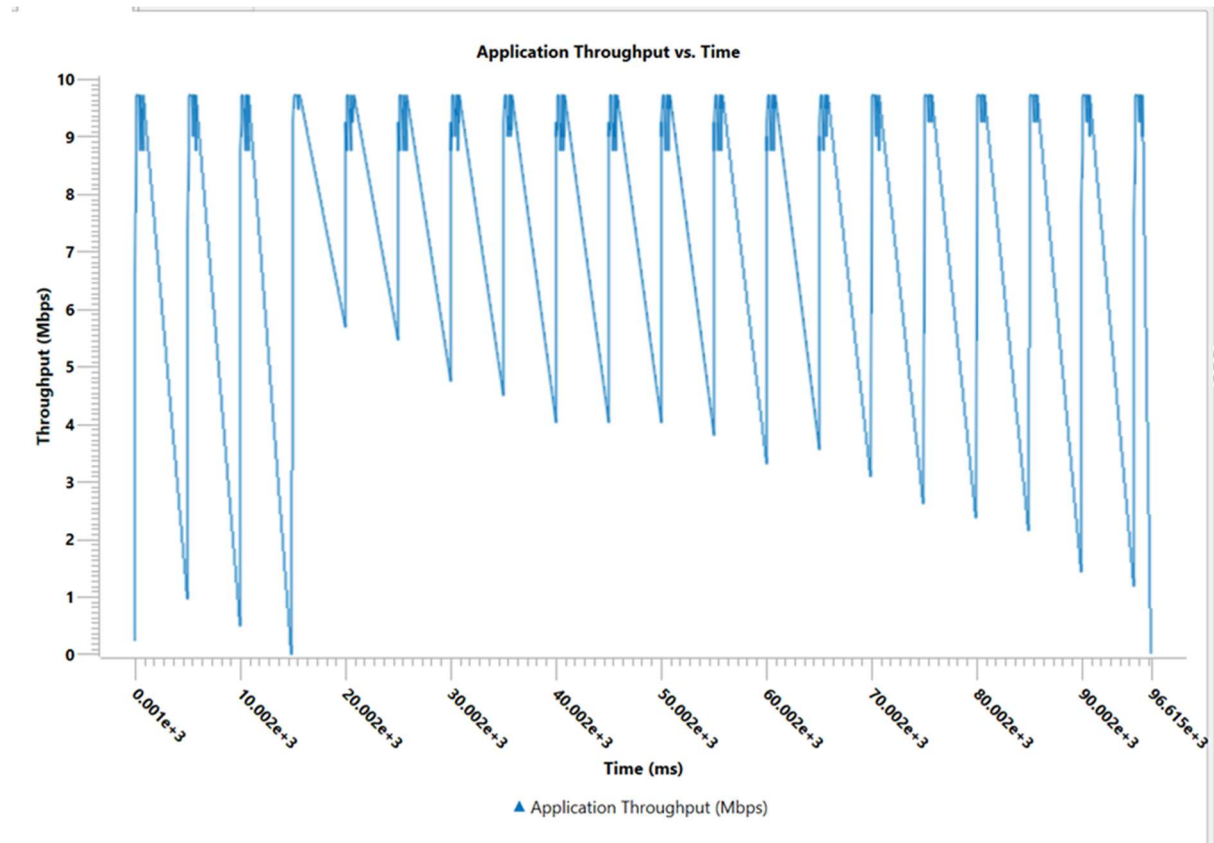
TESTCASE 1:

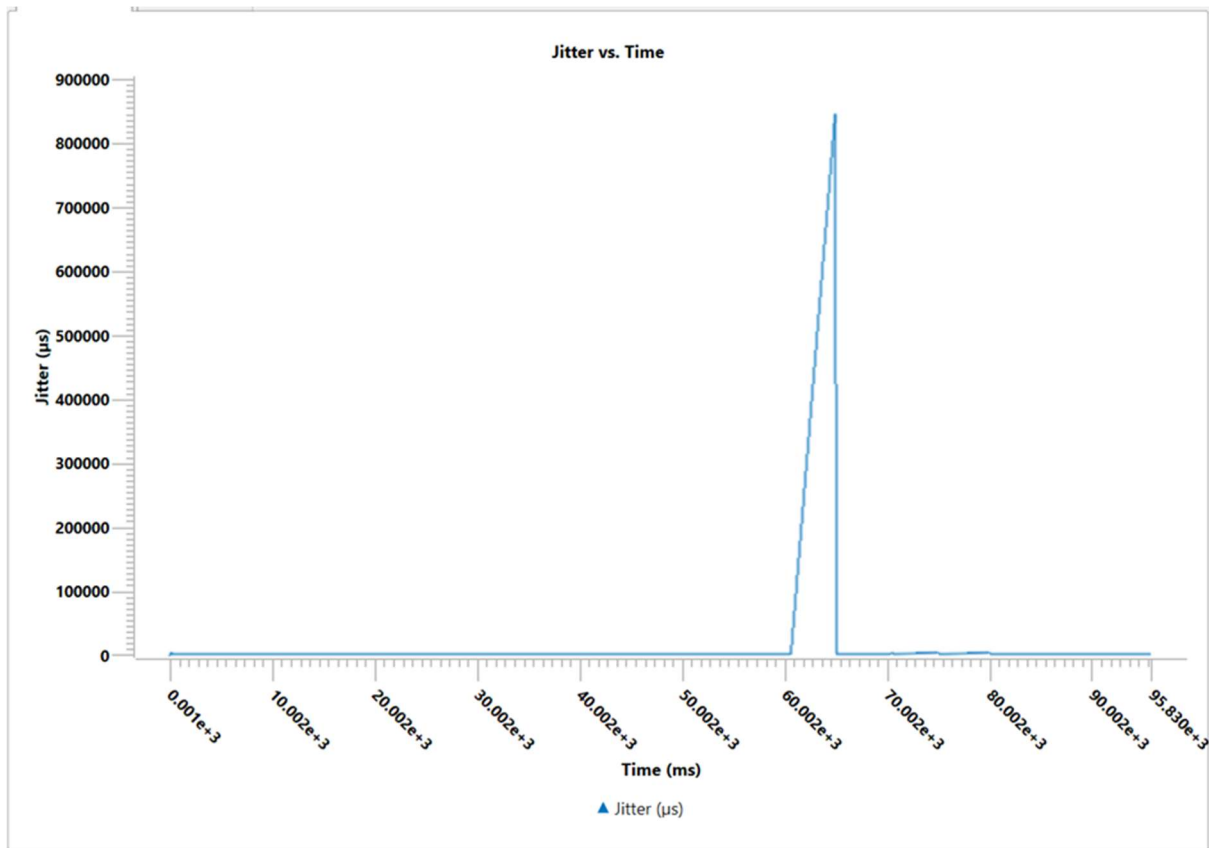


TESTCASE 2:



GRAPHS:





OBSERVATION

As UDP traffic increases, TCP throughput drops sharply due to congestion.

CONCLUSION

TCP and UDP compete for bandwidth; TCP adapts to congestion, while UDP can cause unfairness under heavy load.